

Machine Elements in Mechanical Design

SIXTH EDITION

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MACHINE ELEMENTS IN MECHANICAL DESIGN

Sixth Edition

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Full-Service Management and Composition: R. Sreemeenakshi/SPi Global
Printer/Binder: LSC Communications, Inc.
Cover Printer: Phoenix Color/Hagerstown
Text Font: 10/12 Sabon LT Pro Roman

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Library of Congress Cataloging-in-Publication Data on File

10 9 8 7 6 5 4 3 2 1



ISBN 10: 0-13-444118-4
ISBN 13: 978-0-13-444118-4

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PREFACE

The objective of this book is to provide the concepts, procedures, data, and decision analysis techniques necessary to design machine elements commonly found in mechanical devices and systems. Students completing a course of study using this book should be able to execute original designs for machine elements and integrate the elements into a system composed of several elements.

This process requires a consideration of the performance requirements of an individual element and of the interfaces between elements as they work together to form a system. For example, a gear must be designed to transmit power at a given speed. The design must specify the number of teeth, pitch, tooth form, face width, pitch diameter, material, and method of heat treatment. But the gear design also affects, and is affected by, the mating gear, the shaft carrying the gear, and the environment in which it is to operate. Furthermore, the shaft must be supported by bearings, which must be contained in a housing. Thus, the designer should keep the complete system in mind while designing each individual element. This book will help the student approach design problems in this way.

This text is designed for those interested in practical mechanical design. The emphasis is on the use of readily available materials and processes and appropriate design approaches to achieve a safe, efficient design. It is assumed that the person using the book will be the designer, that is, the person responsible for determining the configuration of a machine or a part of a machine. Where practical, all design equations, data, and procedures needed to make design decisions are specified.

It is expected that students using this book will have a good background in statics, strength of materials, college algebra, and trigonometry. Helpful, but not required, would be knowledge of kinematics, industrial mechanisms, dynamics, materials, and manufacturing processes.

Among the important features of this book are the following:

1. It is designed to be used at the undergraduate level in a first course in machine design.
2. The large list of topics allows the instructor some choice in the design of the course. The format is also appropriate for a two-course sequence and as a reference for mechanical design project courses.
3. Students should be able to extend their efforts into topics not covered in classroom instruction because explanations of principles are straightforward and include many example problems.
4. The practical presentation of the material leads to feasible design decisions and is useful to practicing designers.
5. The text advocates and demonstrates use of computer spreadsheets in cases requiring long, laborious solution procedures. Using spreadsheets allows the designer to make decisions and to modify data at several points within the problem while the computer performs all computations. See Chapter 6 on columns, Chapter 9 on spur gears, Chapter 12 on shafts, Chapter 13 on shrink fits, and Chapter 18 on spring design. Other computer-aided calculation software can also be used.
6. References to other books, standards, and technical papers assist the instructor in presenting alternate approaches or extending the depth or breadth of treatment.
7. Lists of Internet sites pertinent to topics in this book are included at the end of most chapters to assist readers in accessing additional information or data about commercial products.
8. In addition to the emphasis on original design of machine elements, much of the discussion covers commercially available machine elements and devices, since many design projects require an optimum combination of new, uniquely designed parts and purchased components.
9. For some topics the focus is on aiding the designer in selecting commercially available components, such as rolling contact bearings, flexible couplings, ball screws, electric motors, belt drives, chain drives, wire rope, couplings, clutches, and brakes.
10. Computations and problem solutions use both the International System of Units (SI) and the U.S. Customary System (inch-pound-second) approximately equally. The basic reference for the usage of SI units is IEEE/ASTM-SI-10 *American National standard for Metric Practice*. This document is the primary American National Standard on application of the metric system.
11. Extensive appendices are included along with detailed tables in many chapters to help the reader to make real design decisions, using only this text. Several appendix tables feature commercially available structural shapes in both larger and smaller sizes and many in purely metric dimensions are included in this edition to give instructors and students many options for completing design problems.

MECHANICAL DESIGN SOFTWARE

The design of machine elements inherently involves extensive procedures, complex calculations, and many design decisions. Data must be found from numerous charts and tables. Furthermore, design is typically iterative, requiring the designer to try several options for any given element, leading to the repetition of design calculations with new data or new design decisions. This is especially true for complete mechanical devices containing several components as the interfaces between components are considered. Changes to one component often require changes to mating elements. Use of spreadsheets, computational software, and computer-aided mechanical design software can facilitate the design process by performing many of the tasks while leaving the major design decisions to the creativity and judgment of the designer or engineer.

We emphasize that users of computer software must have a solid understanding of the principles of design and stress analysis to ensure that design decisions are based on reliable foundations. We recommend that the software be used only after mastering a given design methodology by careful study and using manual techniques.

The strong movement in the United States and other industrialized countries toward global sourcing of materials and products and the use of multinational design teams makes the use of commercial software highly valuable during the lifelong career of designers and engineers. Furthermore, the specification of commercially available machine components and systems typically involves the use of manufacturers' software built into company Internet sites. This book provides guidance on the use of such sites as an integral part of the machinery design process.

FEATURES OF THE SIXTH EDITION

The practical approach to designing machine elements in the context of complete mechanical designs is retained and refined in this edition. An extensive amount of updating has been accomplished through the inclusion of new photographs of commercially available machine components, new design data for some elements, new or revised standards, new end-of-chapter references, listings of Internet sites, and some completely new elements. Full color has been used for the first time to enhance the visual attractiveness of the book and to highlight prominent features of charts, graphs, and technical illustrations. Numerous, highly detailed, full-color new drawings have been added or have replaced drawings used in previous editions.

The following list summarizes the primary features and the updates.

1. The three-part structure that was introduced in the third edition has been maintained.
 - Part I (Chapters 1–6) focuses on reviewing and upgrading readers' understanding of design philosophies, the principles of strength of materials, the design properties of materials, combined stresses, design for different types of loading, and the analysis and design of columns.
 - Part II (Chapters 7–15) is organized around the concept of the design of a complete power-transmission system, covering some of the primary machine elements such as belt drives, chain drives, wire rope, gears, shafts, keys, couplings, seals, and rolling contact bearings. These topics are tied together to emphasize both their interrelationships and their unique characteristics. Chapter 15, **Completion of the Design of a Power Transmission**, is a guide through detailed design decisions such as the overall layout, detail drawings, tolerances, and fits. Several new, full-color drawings for an example of a gear-type speed reducer have been added to aid students' perception and understanding of how individual machine elements are designed, assembled, and operated together. The representation of the complete single-reduction gear drive at the end of Chapter 15 has been significantly upgraded, aiding students' understanding of how to translate design analysis, decision-making about component details, and commercially available components into a complete assembly.
 - Part III (Chapters 16–22) presents methods of analysis and design of several important machine elements that were not pertinent to the design of a power transmission. These chapters can be covered in any order or can be used as reference material for general design projects. Covered here are plain surface bearings, linear motion elements, fasteners, springs, machine frames, bolted connections, welded joints, electric motors, controls, clutches, and brakes.
2. **The Big Picture, You Are the Designer, and Objectives** features introduced in earlier editions are maintained and refined. Feedback about these features from users, both students and instructors, have been enthusiastically favorable. They help readers to draw on their own experiences and to appreciate what competencies they will acquire from the study of each chapter. Constructivist theories of learning espouse this approach.
3. Lists of Internet sites and printed references have been updated and edited in every chapter. Many new entries have been added. The extensive lists of such resources are useful to students, instructors, and practicing engineers to extend their understanding of concepts beyond this book and to access the huge

potential of the Internet as a source of information about practical design methods and commercially available products.

4. Some of the new or updated topics from individual chapters are summarized here.

- In Chapter 1, **The Nature of Mechanical Design**, first ten figures showing a variety of mechanical devices and machinery have been replaced with new, full-color images to enhance students' perceptions of the details of many types of equipment. Two of these new images show production machinery designed by one of the new coauthors of this book.
- Chapter 2, **Materials in Mechanical Design**, continues to emphasize the specification and use of appropriate materials, building on prior courses in metallurgy, materials, and processes. Extensive tables listing materials commonly used in commercially available shapes are included. To serve the global nature of machine design, an extensive table of designations for steel and aluminum alloys from several countries is included. Designations for steel alloys continue to use the SAE numbering system. The discussion of heat treating of steels continues to focus on quenching and tempering along with case hardening to give students an appreciation of the wide range of properties that any given material can have and the importance of being able to specify pertinent heat treatment requirements. Descriptions of white iron, powder metals, aluminum casting and forging alloys, magnesium, nickel-based alloys, titanium alloys, and brasses and bronzes are included. The extensive discussion of advanced engineering composites includes SI data, nanocomposites, and design approaches, continuing to provide students with basic concepts that can lead to novel applications of composite materials to machine design. Materials selection using decision analysis techniques has been refined.
- Chapter 3, **Stress and Deformation Analysis**, has been reorganized with some section titles revised, bringing an improved order of coverage. The objective of the update is to clarify how the external loading, such as direct normal force, direction shear force, torsion/torque, and bending moment can produce normal and shear stresses on a stress element.
- Graphs of stress concentration factors have been returned to the Appendix, allowing students to apply them in most problem-solving exercises in this book. However, information about other print and easily-accessible Internet sources for K_t values remain, giving instructors and students the opportunity to apply a wider scope of design data.
- Chapter 4, **Combined Stresses and Stress Transformation**, has been revised to show that a stress element is always 3-dimensional (3D). Resulting from some loading condition, the stresses on a 3D element, however, can be in 1D or 2D stress state. That is, the values of stress components in certain direction(s) can be zero. This concept is presented to assist readers in analyzing 3D combined loading and combined stress problems. A major change in this chapter is that, while the Mohr's circle technique is used for 1D or 2D stress transformation, the resulting stress element is presented in 3D, having one or two principal stresses equal to zero. The 3D approach can help readers to visualize the stress state of a point (a stress element) at the location of interest.
- The contents of Chapter 5, **Design for Different Types of Loading**, have been reorganized and a brief discussion of failure theories has been added, extending the revisions discussed for Chapters 3 and 4. The design methods for static loading and cyclic loading are now clearly identified in different sections. All stress elements in the Design Examples are 3D elements, while recognizing that some elements are in a 1D or 2D stress state. For failure prediction, a unified approach based on the evaluation of principal stresses against material properties is presented. The 3D approach is also used in mean and alternating stresses calculation when dealing with fatigue failure in cyclic loading condition. Continuing from the 5th edition, are discussions of endurance strength, recommended design and processing approaches under fatigue load, the Smith Diagram approach for showing the effect of mean stress on fatigue, and the damage accumulation method for varying stress amplitudes.
- In Chapter 7, **Belt Drives, Chain Drives, and Wire Rope**, significant new material on synchronous belt drive designs in both SI and U.S. units has been added. Common metric sizes for V-belts, synchronous belts, chains, and sprockets are included. The new section on wire rope complements the former parts of this chapter with information that can be applied to lifting equipment and industrial machinery for which flexible tensile elements are needed.
- Chapter 8, **Kinematics of Gears**, continues to emphasize the geometry of U. S. and metric module-type gearing and has an integrated discussion of spur, helical, bevel, and wormgearing. A useful table for calculating key geometric features of gears and gear teeth aids problem solving and design decisions. Discussions of velocity ratios, train values, and devising gear trains have been refined and new, detailed, color drawings are included.

- Chapter 9, **Spur Gear Design**, continues to be refined in its use of AGMA standards along with the metric module system. The arrangement of sections has been modified for smoother coverage of the various aspects of gear design. Additional example problems illustrate different approaches to the design process. Topics covering gear lubricants and typical viscosity grades are included.
- Chapter 10, **Helical Gears, Bevel Gears, and Wormgearing**, has been updated along similar lines as discussed for Chapter 9 on Spur Gear Design.
- In Chapter 11, **Keys, Couplings, and Seals**, new information is provided for selecting flexible couplings and universal joints.
- In Chapter 12, **Shaft Design**, the highly regarded procedure for the design of a shaft has been continued. Coverage of the torque capacity of selected flexible shaft sizes continues.
- In Chapter 14, **Rolling Contact Bearings**, the bearing selection procedure has been closely tied to the use of manufacturers' data and the specific procedures outlined on their Internet sites, listed at the end of the chapter. This permits the use of a wide variety of sources and types of bearings as is done in practical mechanical design. Sample data are included in the chapter to introduce students to the variables involved in bearing selection and the types of analysis required to specify optimal bearings. An extensive discussion of bearing materials is included for steels, ceramics, Monel, titanium/nickel alloys, and plastics to emphasize the importance of specifying materials that meet application requirements.
- Chapter 16, **Plain Surface Bearings**, includes sample data on pV factors for boundary-lubricated bearings and common lubricants, along with the analysis of plain bearing performance under oscillating motion. Coverage of topics such as hydrodynamic and hydrostatic bearings continues. An intriguing new example of the application of boundary lubrication, called the Kugel Fountain, has been added.
- In Chapter 17, **Linear Motion Elements**, new information about high-speed linear actuators has been added to the discussion of power screws and ball screw drives.
- Chapter 18 on **Springs**, Chapter 19 on **Fasteners**, and Chapter 20, **Frames, Bolted Connections, and Welded Joints** provide useful information about components and analysis techniques used in many types of machinery.
- Chapter 21 **Electric Motors and Controls**, and Chapter 22, **Motion Control: Clutches and Brakes**, assist the mechanical designer in specifying electrical drive systems and electrical and mechanical controls for a wide variety of applications.

- **The Appendix** has an extensive set of tables for material properties of steels, cast irons, aluminum alloys, zinc and magnesium alloys, plastics, nickel-based alloys, titanium alloys, bronzes, brasses, and other copper alloys. Several tables of data are included for section properties of commercially available shapes in larger and smaller sizes and in pure metric dimensions to provide a wide array of choices for problem-solving and design. Appendixes for beam deflection formulas, conversion factors, and hardness assist students as they study multiple chapters. Ten charts for stress concentration factors have been returned to the book in a revised order that is related to the manner of loading; tension, bending, and torsion.

INTRODUCING TWO NEW CO-AUTHORS:

For the first five editions of this book, the sole author was Robert L. Mott. For this new 6th edition, two outstanding co-authors have contributed to a great extent in updating and upgrading the content, and enhancing the appearance of the book. Their brief biographies are mentioned below. For those using this book and who may not know Professor Mott, his brief biography follows:

Robert L. Mott is Professor Emeritus of Engineering Technology at the University of Dayton. He is a member of ASEE, SME, and ASME. He is a Fellow of ASEE and a recipient of the ASEE James H. McGraw Award and the Archie Higdon Distinguished Educator Award from the Mechanics Division. He is a recipient of the SME Education Award for his contributions to manufacturing education. He holds the Bachelor of Mechanical Engineering degree from General Motors Institute (Now Kettering University) and the Master of Science in Mechanical Engineering from Purdue University. He has authored three textbooks; *Applied Fluid Mechanics* 7th ed. (2015) and *Machine Elements in Mechanical Design* 6th ed. (2018), published by Pearson; *Applied Strength of Materials* 6th ed. (2017) published by CRC Press. His work experience includes serving as a research engineer for General Motors Corporation, consulting for industrial clients, working for the University of Dayton Research Institute (UDRI), leading the Center for Advanced Manufacturing for UDRI, and serving as an expert witness for accident analysis cases for industrial and automotive accidents. He also served for 12 years as one of the senior personnel for the NSF-sponsored National Center for Manufacturing Education based in Dayton, Ohio.

Edward M. Vavrek is an Associate Professor in Mechanical Engineering Technology at Purdue University Northwest, located at the Westville, IN campus, an extension of Purdue University. He is a member of AGMA, ASME, and ASEE. He received his Bachelor of Science in

Mechanical Engineering from Purdue University Calumet, Masters in Business Administration from Indiana University Northwest, and Masters in Mechanical and Aeronautical Engineering from the Illinois Institute of Technology. He has significant industrial experience in design and development of machinery, using SolidWorks and Inventor, within the printing/converting, shipbuilding, railroad, steel mill, and automotive industries. He has presented multiple papers on his software developed for the area of machine design. He holds one U.S. patent. He also does extensive private consulting in mechanical design that is highly relevant to the content of this book.

Dr. Jyhwen Wang, Ph.D. is a Professor with dual appointment in the departments of Engineering Technology and Industrial Distribution and Mechanical Engineering at Texas A&M University in College Station, TX. He holds the degrees of Ph.D. in Mechanical Engineering and Master of Engineering in Manufacturing

Engineering from Northwestern University in Evansville, IL, the M.S. in Industrial Engineering and Operations Research from Syracuse University in Syracuse, NY, and the B.S. in Industrial Engineering from Tung-hai University in Taichung, Taiwan. He has significant industrial experience with Weirton Steel Corporation in Weirton, West Virginia along with consulting for several organizations. He has participated in funded research and education projects as PI or Co-PI. He is a Fellow of the American Society of Mechanical Engineers and the Society of Manufacturing Engineers. Professional society memberships include ASME, ASEE, SME, NAMRI/SME (North American Manufacturing Research Institute), and NADDRG (North American Deep Drawing Research Group). He has written book sections for *Manufacturing Processes for Engineering Materials*, (2003) and *Manufacturing Engineering and Technology*, (2001) by Kalpakjian and Schmid published by Pearson.

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ACKNOWLEDGMENTS

Our appreciation is extended to all those who provided helpful suggestions for improvements to this book. We thank the editorial staff of Pearson, those who provided illustrations, and the many users of the book, both instructors and students, with whom we have had discussions. Special appreciation goes to our colleagues at the University of Dayton, Purdue University, and Texas A&M University. We would like to thank Amit Banerjee, The Pennsylvania State University; Michael DeVore, Cincinnati State Technical and Community College; Dexter Hulse, University of Cincinnati; Scott Kessler, Colorado Mesa University;

and Zhongming Liang, Indiana Purdue University Fort Wayne, for their helpful reviews of this revision. We also thank those who provided thoughtful reviews of prior editions. We especially thank our students—past and present—for their encouragement and positive feedback about this book. All three co-authors extend sincere gratitude to our wives, children, and parents who provided unwavering support, patience, and inspiration as we prepared the new 6th edition of this book.

Robert L. Mott, Edward M. Vavrek, Jyhwen Wang

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PRINCIPLES OF DESIGN AND STRESS ANALYSIS

As you complete the first six chapters of this book, you will gain an understanding of design philosophies, and you will build on earlier-learned principles of strength of materials, materials science, and manufacturing processes. The competencies gained from these chapters are useful throughout the book and in general machine design or product design projects.

Chapter 1: The Nature of Mechanical Design helps you see the big picture of the process of mechanical design. Several examples are shown from different industry sectors: Consumer products, manufacturing systems, construction equipment, agricultural equipment, transportation equipment, ships, and space systems. The responsibilities of designers are discussed, along with an illustration of the iterative nature of the design process. Units and conversions structural shapes, screw threads, and preferred basic sizes complete the chapter.

Chapter 2: Materials in Mechanical Design emphasizes the design properties of materials. Much of this chapter is probably review for you, but it is presented here to emphasize the importance of material selection to the design process and to explain the data for materials presented in the Appendices.

Chapter 3: Stress and Deformation Analysis is a review of the basic principles of stress and deflection analysis. It is essential that you understand the basic concepts summarized here before proceeding with later material. Reviewed are direct tensile, compressive, and shearing stresses; bending stresses; and torsional shear stresses.

Chapter 4: Combined Stresses and Stress Transformations is important because many general design problems and the design of machine elements covered in later chapters of the book involve combined stresses. You may have covered these topics in a course in strength of materials.

Chapter 5: Design for Different Types of Loading is an in-depth discussion of design factors, fatigue, and many of the details of stress analysis as used in this book.

Chapter 6: Columns discusses the long, slender, axially loaded members that tend to fail by buckling rather than by exceeding the yield, ultimate, or shear stress of the material. Special design and analysis methods are reviewed here.

THE NATURE OF MECHANICAL DESIGN

The Big Picture

You Are the Designer

- 1-1 Objectives of This Chapter
- 1-2 The Design Process
- 1-3 Skills Needed in Mechanical Design
- 1-4 Functions, Design Requirements, and Evaluation Criteria
- 1-5 Example of the Integration of Machine Elements into a Mechanical Design
- 1-6 Computational Aids
- 1-7 Design Calculations
- 1-8 Preferred Basic Sizes, Screw Threads, and Standard Shapes
- 1-9 Unit Systems
- 1-10 Distinction among Weight, Force, and Mass

THE BIG PICTURE

The Nature of Mechanical Design

Discussion Map

- To design mechanical components and devices, you must be competent in the design of individual elements that comprise the system.
- But you must also be able to integrate several components and devices into a coordinated, robust system that meets your customer's needs.

Discover

Think, now, about the many fields in which you can use mechanical design:

What are some of the products of those fields?

What kinds of materials are used in the products?

What are some of the unique features of the products?

How were the components made?

How were the parts of the products assembled?

Consider consumer products, construction equipment, agricultural machinery, manufacturing systems, and transportation systems on the land, in the air, in space, and on and under water.

In this book, you will find the tools to learn the principles of **Machine Elements in Mechanical Design**.

Design of machine elements is an integral part of the larger and more general field of mechanical design. Designers and design engineers create devices or systems to satisfy specific needs. Mechanical devices typically involve moving parts that transmit power and accomplish specific patterns of motion. Mechanical systems are composed of several mechanical devices.

Therefore, to design mechanical devices and systems, you must be competent in the design of individual machine elements that comprise the system. But you must also be able to integrate several components and

devices into a coordinated, robust system that meets your customer's needs. From this logic comes the name of this book, *Machine Elements in Mechanical Design*.

Think about the many fields in which you can use mechanical design. Discuss these fields with your instructor and with your colleagues who are studying with you. Talk with people who are doing mechanical design in local industries. Try to visit their companies if possible, or meet designers and design engineers at meetings of professional societies. Consider the following fields where mechanical products are designed and produced.

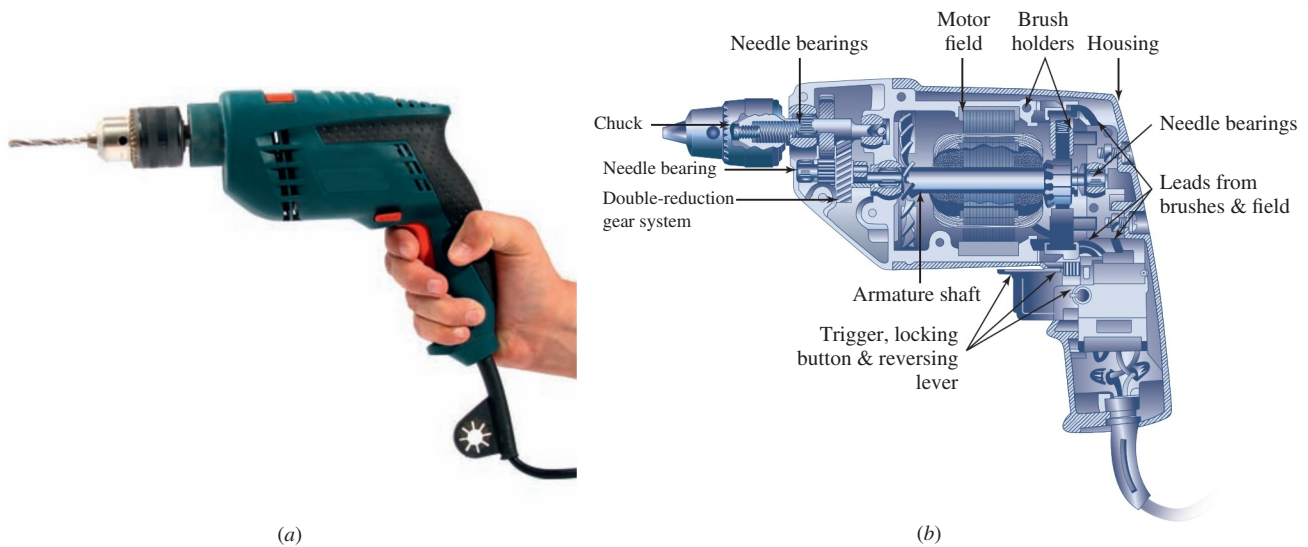
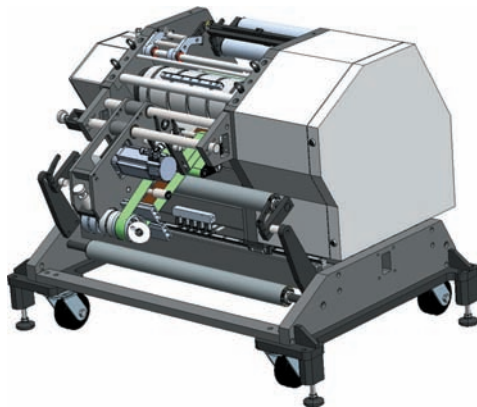


FIGURE 1-1 (a) Hand-held power drill (b) Cutaway view of a hand drill

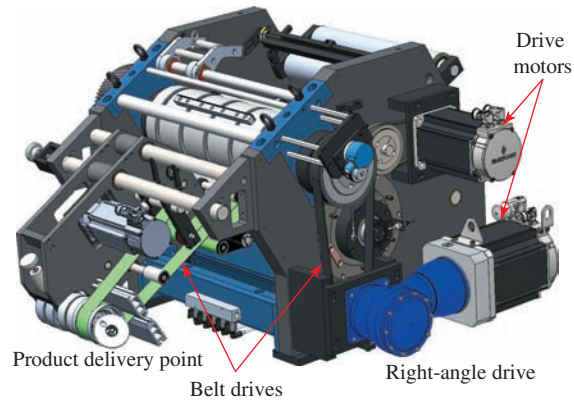
- **Consumer products:** Household appliances (can openers, food processors, mixers, toasters, vacuum cleaners, clothes washers), lawn mowers, chain saws, power tools, garage door openers, air-conditioning systems, and many others. See Figures 1-1 and 1-2 for a few examples of commercially available products.
- **Manufacturing systems:** Material handling devices, conveyors, cranes, transfer devices, industrial robots, machine tools, automated assembly systems, special-purpose processing systems, forklift trucks, and packaging equipment. See Figures 1-3 and 1-4.
- **Construction equipment:** Tractors with front-end loaders or backhoes, cranes, power shovels, earthmovers, graders, dump trucks, road pavers, concrete mixers, powered nailers and staplers, compressors, and many others. See Figures 1-5 and 1-6.
- **Agricultural equipment:** Tractors, harvesters (for corn, wheat, tomatoes, cotton, fruit, and many other crops), rakes, hay balers, plows, disc harrows, cultivators, and conveyors. See Figures 1-6, 1-7, and 1-8.
- **Transportation equipment:** (a) Automobiles, trucks, and buses, which include hundreds of mechanical devices such as suspension components (springs, shock absorbers, and struts); door and window operators; windshield wiper mechanisms; steering systems; hood and trunk latches and hinges; clutch and braking systems; transmissions; driveshafts; seat adjusters; and numerous parts of the engine systems. (b) Aircraft, which include retractable landing gear, flap and rudder actuators, cargo-handling devices, seat reclining mechanisms, dozens of latches, structural components, and door operators. See Figures 1-9 and 1-10.



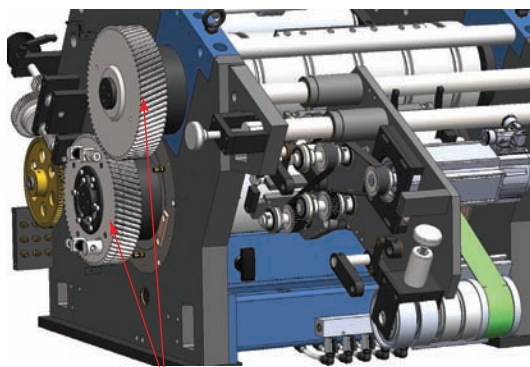
FIGURE 1-2 Chain saw
(Shutterstock)



(a) Pictorial view with enclosures over mechanisms

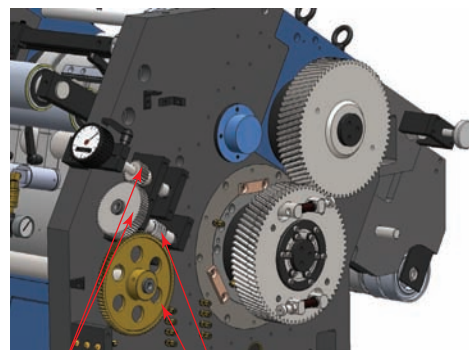


(b) Covers removed to show right side drive systems



Anti-backlash helical gear set

(c) Left side, viewed from front of machine

Spur gear pair
Worm and wormgear

(d) Closer view of gear drives from left-rear

FIGURE 1-3 Closed end mailer for the printing industry

Created by Edward M. Vavrek for The Lettershop Group, Leeks, UK

- **Ships:** Winches to haul up the anchor, cargo-handling cranes, rotating radar antennas, rudder steering gear, drive gearing and driveshafts, and the numerous sensors and controls for operating on-board systems.
- **Space systems:** Satellite systems, the space shuttle, the space station, and launch systems, which contain numerous mechanical systems such as devices to deploy antennas, hatches, docking systems, robotic arms, vibration control devices, devices to secure cargo, positioning devices for instruments, actuators for thrusters, and propulsion systems.

How many examples of mechanical devices and systems can you add to these lists?

What are some of the unique features of the products in these fields?

What kinds of mechanisms are included?

What kinds of materials are used in the products?

How were the components made?

How were the parts assembled into the complete products?

In this book, you will find the tools to learn the principles of *Machine Elements in Mechanical Design*. In the introduction to each chapter, we include a brief scenario called *You Are the Designer*. The purpose of these scenarios is to stimulate your thinking about the material presented in the chapter and to show examples of realistic situations in which you may apply it.

Let's consider Figures 1-3 and 1-4 more closely to show specific examples of how coverage of machine elements relates directly to mechanical design.

Figure 1-3 shows a *closed end mailer* for the printing industry. It takes in bulk paper, forms it, cuts it, and delivers it to the user. Clearly shown are electric motor drives, a right-angle gearbox, a belt drive to rotate rollers, a helical gear pair, a spur gear pair, a worm/wormgear set, bearings, and several other types

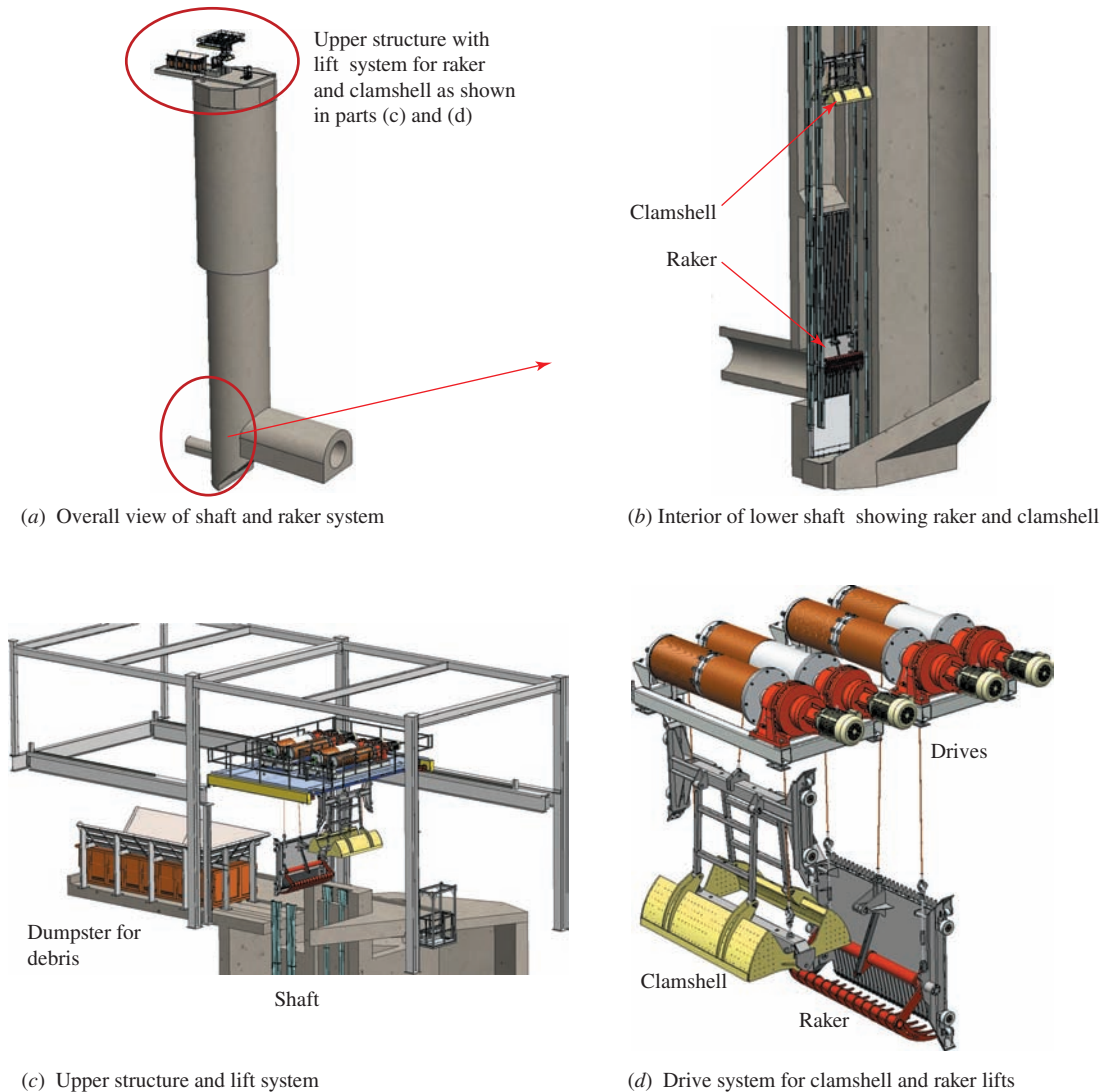


FIGURE 1-4 Raker and clamshell for a deep rock tunnel connector pumping station

Created by Edward M. Vavrek for Fairfield Service Company, Michigan City, IN

FIGURE 1-5 Construction crane on a building site
(Shutterstock)





FIGURE 1-6 Construction backhoe and front-end loader (Shutterstock)



FIGURE 1-7 Corn harvester on a farm (Shutterstock)



FIGURE 1-8 Heavy duty tractor for farm, highway construction, and commercial applications (Shutterstock)

FIGURE 1–9 Aircraft showing open door and steps
(Shutterstock)



FIGURE 1–10 Landing gear for a large aircraft
(Shutterstock)



of mechanisms. Scan the chapter and section titles for this book to see how these topics are presented.

Figure 1–4 shows a huge system called *deep rock tunnel connector pumping system*. The vertical shaft is over 250 ft deep and it is part of a water storage system for a major city. Of most interest to this book is the mechanical drive system on the upper part of the shaft that lifts a raker system that separates

debris from a huge screen in the water flow path. The debris falls to the bottom of the shaft and the clam-shell device picks it up, raises it to the top of the shaft and then transports it horizontally to dispose it into a specially designed dumpster. The drive systems, wire-rope lifting systems, actuating mechanisms, transfer system, and other components are highly relevant to the topics presented in this book.

YOU ARE THE DESIGNER

Consider, now, that you are the designer responsible for the design of a new consumer product, such as the hand drill for a home workshop shown in Figure 1–1. What kind of technical preparation would you need to complete the design? What steps would you follow? What information would you need? How would you show, by calculation, that the design is safe and that the product will perform its desired function?

The general answers to these questions are presented in this chapter. As you complete the study of this book, you will learn about many design techniques that will aid in your design of a wide variety of machine elements. You will also learn how to integrate several machine elements into a mechanical system by considering the relationships between and among elements. ■

1-1 OBJECTIVES OF THIS CHAPTER

After completing this chapter, you will be able to:

1. Recognize examples of mechanical systems in which the application of the principles discussed in this book is necessary to complete their design.
2. List what design skills are required to perform competent mechanical design.
3. Describe the importance of integrating individual machine elements into a more comprehensive mechanical system.
4. Describe the main elements of the *product realization process*.
5. Write statements of *functions* and *design requirements* for mechanical devices.
6. Establish a set of criteria for evaluating proposed designs.
7. Work with appropriate units in mechanical design calculations both in the U.S. Customary Unit System and in SI metric units.
8. Distinguish between *force* and *mass*, and express them properly in both unit systems.
9. Present design calculations in a professional, neat, and orderly manner that can be understood and evaluated by others knowledgeable in the field of machine design.
10. Become acquainted with section properties of commercially available structural shapes and other tables of data in the Appendix of this book to aid in performing design and analysis tasks throughout the book.

1-2 THE DESIGN PROCESS

The ultimate objective of mechanical design is to produce a useful product that satisfies the needs of a customer and that is safe, efficient, reliable, economical, and practical to manufacture. Think broadly when answering the question, “Who is the customer for the product or system I am about to design?” Consider the following scenarios:

- **You are designing a can opener for the home market.** The ultimate customer is the person who will purchase the can opener and use it in the kitchen of a home. Other customers may include the designer of the packaging for the opener, the manufacturing staff who must produce the opener economically, and service personnel who repair the unit.
- **You are designing a piece of production machinery for a manufacturing operation.** The customers include the manufacturing engineer who is responsible for the production operation, the operator of the machine, the staff who install the machine, and the maintenance personnel who must service the machine to keep it in good running order.
- **You are designing a powered system to open a large door on a passenger aircraft.** The customers include the person who must operate the door in normal service or in emergencies, the people who must pass through the door during use, the personnel who manufacture the opener, the installers, the aircraft structure designers who must accommodate the loads produced by the opener during flight and during operation, the service technicians who maintain the system, and the interior designers who must shield the opener during use while allowing access for installation and maintenance.

It is essential that you know the desires and expectations of all customers before beginning product design. Marketing professionals are often employed to manage the definition of customer expectations, but designers will likely work with them as a part of a product development team.

Numerous approaches are available that guide designers through the complete process of product design and methods for creating new, innovative products. Some are oriented toward large complex products such as aircraft, automobiles, and multifunction machine tools. It is advisable for a company to select one method that is suitable to their particular style of products or to create one that meets their special needs. The following discussion identifies the salient features of some of the approaches and the listed references and Internet sites provide more details. Some of the listed methods are applied in combination.

- **Axiomatic design.** See References 14, 15, and 18 and Internet site 8. Axiomatic design methods implement a process where developers think functionally first, followed by the innovative creation of the physical embodiment of a product to meet customer requirements along with the process needed to produce the product.
- **Quality function deployment (QFD).** See Reference 8 and Internet sites 9 and 10. QFD is a quality system that espouses understanding customer requirements and uses quality systems thinking to maximize positive quality that adds value. The process also includes use of the “House of Quality” matrix described in Reference 8.
- **Design for six sigma (DFSS).** See References 18–20 and Internet sites 11 and 16. The objective of Six Sigma Quality is to reduce output variation that will result in no more than 3.4 defective parts per million (PPM). The term *six sigma* or 6σ refers to a distribution of performance measures, in which products produced are within upper and lower specification limits that are six process standard deviations from the mean.
- **TRIZ.** See References 21–23 and Internet sites 12–15. TRIZ is an acronym for a Russian phrase that translates into English as “Theory of Inventive Problem

Solving.” Developed in 1946 in Russia by Genrich Altshuller and colleagues, the process is applied throughout the world to create and to improve products, services, and systems. TRIZ is a problem-solving method based on logic and data, not intuition, which accelerates the project team’s ability to solve problems creatively.

- **Total design.** See Reference 13. An integrated approach to product engineering using a systematic and disciplined process to create innovative products that satisfy customer needs.
- **The engineering design process—embodiment design.** See Reference 26. A comprehensive process involving need identification, concept selection, decision making, detail design, modeling and simulation, design for manufacturing, robust design, and several other elements.
- **Failure modes and effects analysis (FMEA).** See Reference 24 and Internet site 17. An analysis technique which facilitates the identification of potential problems in the design of a product by examining the effects of lower level failures. Recommended actions or compensating provisions are made to reduce the likelihood of the problem occurring or mitigating the risk if problems do occur. The process evolved from military and NASA procedures designed to enhance the reliability of products and systems. For many years, MIL STD 1629A defined accepted FMEA methods used in military and commercial industry. Even though that standard was cancelled, it remains the basis for much of the current work related to FMEA. MIL-Handbook-502A Product Support Analysis is now widely used. The prevalent standards in the automotive and aerospace industries are SAE J1739 and SAE TA-STD-0017, Product Support Analysis.
- **Product design for manufacture and assembly.** See Reference 27. A product design methodology with a heavy emphasis on how the components and the assembled product are to be manufactured to achieve a low-cost, high-quality design. Included are design for die casting, forging, powder metal processing, sheet metalworking, machining, injection molding, and many other processes.

It is also important to consider how the design process fits with all functions that must happen to deliver a satisfactory product to the customer and to service the product throughout its life cycle. In fact, it is important to consider how the product will be disposed of after it has served its useful life. The total of all such functions that affect the product is sometimes called the *product realization process* or *PRP*. (See References 3 and 10.) Some of the factors included in PRP are as follows:

- Marketing functions to assess customer requirements.
- Research to determine the available technology that can reasonably be used in the product.
- Availability of materials and components that can be incorporated into the product.
- Product design and development.
- Performance testing.
- Documentation of the design.
- Vendor relationships and purchasing functions.
- Consideration of global sourcing of materials and global marketing.
- Work-force skills.
- Physical plant and facilities available.
- Capability of manufacturing systems.
- Production planning and control of production systems.
- Production support systems and personnel.
- Quality systems requirements.
- Operation and maintenance of the physical plant.
- Distribution systems to get products to the customer.
- Sales operations and time schedules.
- Cost targets and other competitive issues.
- Customer service requirements.
- Environmental concerns during manufacture, operation, and disposal of the product.
- Legal requirements.
- Availability of financial capital.

Can you add to this list?

You should be able to see that the design of a product is but one part of a comprehensive process. In this book, we will focus more carefully on the design process itself, but the producibility of your designs must always be considered. This simultaneous consideration of product design and manufacturing process design is often called *concurrent engineering*. Note that this process is a subset of the larger list given previously for the product realization process. Other major books discussing general approaches to mechanical design are listed as References 6, 7, and 12–26.

1–3 SKILLS NEEDED IN MECHANICAL DESIGN

Product engineers and mechanical designers use a wide range of skills and knowledge in their daily work, including the following:

1. Sketching, technical drawing, and 2D and 3D computer-aided design.
2. Properties of materials, materials processing, and manufacturing processes.
3. Applications of chemistry such as corrosion protection, plating, and painting.
4. Statics, dynamics, strength of materials, kinematics, and mechanisms.

5. Oral communication, listening, technical writing, and teamwork skills.
6. Fluid mechanics, thermodynamics, and heat transfer.
7. Fluid power, the fundamentals of electrical phenomena, and industrial controls.
8. Experimental design, performance testing of materials and mechanical systems, and use of computer-aided engineering software.
9. Creativity, problem solving, and project management.
10. Stress analysis.
11. Specialized knowledge of the behavior of machine elements such as gears, belt drives, chain drives, shafts, bearings, keys, splines, couplings, seals, springs, connections (bolted, riveted, welded, adhesive), electric motors, linear motion devices, clutches, and brakes.

It is expected that you will have acquired a high level of competence in items 1–5 in this list prior to beginning the study of this text. The competencies in items 6–8 are typically acquired in other courses of study either before, concurrently, or after the study of design of machine elements. Item 9 represents skills that are developed continuously throughout your academic study and through experience. Studying this book will help you acquire significant knowledge and skills for the topics listed in items 10 and 11.

1–4 FUNCTIONS, DESIGN REQUIREMENTS, AND EVALUATION CRITERIA

Section 1–2 emphasized the importance of carefully identifying the needs and expectations of the customer prior to beginning the design of a mechanical device. You can formulate these by producing clear, complete statements of *functions*, *design requirements*, and *evaluation criteria*:

- **Functions** tell what the device must do, using general, nonquantitative statements that employ action phrases such as *to support a load*, *to lift a crate*, *to transmit power*, or *to hold two structural members together*.
- **Design requirements** are detailed, usually quantitative statements of *expected performance levels*, *environmental conditions in which the device must operate*, *limitations on space or weight*, or *available materials and components that may be used*.
- **Evaluation criteria** are statements of *desirable qualitative characteristics* of a design that assist the designer in deciding which alternative design is optimum—that is, the design that maximizes benefits while minimizing disadvantages.

Together these elements can be called the *specifications* for the design.

Most designs progress through a cycle of activities as outlined in Figure 1–11. You should typically propose more than one possible alternative design concept. This is where creativity is exercised to produce truly novel designs. Each design concept must satisfy the functions and design requirements. A critical evaluation of the desirable features, advantages, and disadvantages of each design concept should be completed. Then a rational decision analysis technique should use the evaluation criteria to decide which design concept is the optimum and, therefore, should be produced. See References 25 and 28 and Internet site 18.

The final block in the design flowchart is the detailed design, and the primary focus of this book is on that part of the overall design process. It is important to recognize that a significant amount of activity precedes the detailed design.

Note in Figure 1–11, that in many design projects there are reasons for returning to an earlier stage of the process outlined in Figure 1–11, based on discoveries made later in the process. After moving forward with proposing design concepts, you may discover that initial specifications or design requirements were unreasonable or that something was missing. Then you would return to Phase I to adjust the specifications. This process is called *iteration* and it is very typical in design projects. Other iterative steps are implied in the figure as well.

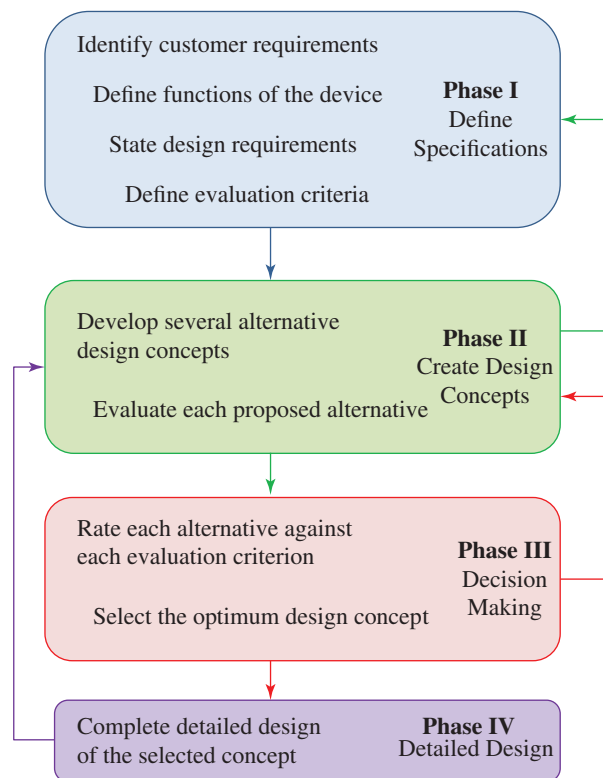


FIGURE 1–11 Steps in the product development and design process

Example of Functions, Design Requirements, and Evaluation Criteria

Consider that you are the designer of a speed reducer that is part of the power transmission for a small tractor. The tractor's engine operates at a fairly high speed, while the drive for the wheels must rotate more slowly and transmit a higher torque than is available at the output of the engine.

To begin the design process, let us list the *functions* of the speed reducer. What is it supposed to do? Some answers to this question are as follows:

Functions

1. To receive power from the tractor's engine through a rotating shaft.
2. To transmit the power through machine elements that reduce the rotational speed to a desired value.
3. To deliver the power at the lower speed to an output shaft that ultimately drives the wheels of the tractor.

Now the *design requirements* should be stated. The following list is hypothetical, but if you were on the design team for the tractor, you would be able to identify such requirements from your own experience and ingenuity and/or by consultation with fellow designers, marketing staff, manufacturing engineers, service personnel, suppliers, and customers.

The product realization process calls for personnel from all of these functions to be involved from the earliest stages of design.

Design Requirements

1. The reducer must transmit 15.0 hp.
2. The input is from a two-cylinder gasoline engine with a rotational speed of 2000 rpm.
3. The output delivers the power at a rotational speed in the range of 290 to 295 rpm.
4. A mechanical efficiency of greater than 95% is desirable.
5. The minimum output torque capacity of the reducer should be 3050 pound-inches (lb · in).
6. The reducer output is connected to the driveshaft for the wheels of a farm tractor. Moderate shock will be encountered.
7. The input and output shafts must be in-line.
8. The reducer is to be fastened to a rigid steel frame of the tractor.
9. Small size is desirable. The reducer must fit in a space no larger than 20 in × 20 in, with a maximum height of 24 in.
10. The tractor is expected to operate 8 hours (h) per day, 5 days per week, with a design life of 10 years.
11. The reducer must be protected from the weather and must be capable of operating anywhere in the United States at temperatures ranging from 0 to 130°F.

12. Flexible couplings will be used on the input and output shafts to prohibit axial and bending loads from being transmitted to the reducer.
13. The production quantity is 10 000 units per year.
14. A moderate cost is critical to successful marketing.
15. All government and industry safety standards must be met.

Careful preparation of function statements and design requirements will ensure that the design effort is focused on the desired results. Much time and money can be wasted on designs that, although technically sound, do not meet design requirements. Design requirements should include everything that is needed, but at the same time they should offer ample opportunity for innovation.

Evaluation criteria should be developed by all members of a product development team to ensure that the interests of all concerned parties are considered. Often weights are assigned to the criteria to reflect their relative importance.

Safety must always be the paramount criterion. Different design concepts may have varying levels of inherent safety in addition to meeting stated safety requirements as noted in the design requirements list. Designers and engineers are legally liable if a person is injured because of a design error. You must consider any reasonably foreseeable uses of the device and ensure safety of those operating it or those who may be close by.

Achieving a high overall performance should also be a high priority. Certain design concepts may have desirable features not present on others.

The remaining criteria should reflect the special needs of a particular project. The following list gives examples of possible evaluation criteria for the small tractor.

Evaluation Criteria

1. Safety (the relative inherent safety over and above stated requirements).
2. Performance (the degree to which the design concept exceeds requirements).
3. Ease of manufacture.
4. Ease of service or replacement of components.
5. Ease of operation.
6. Low initial cost.
7. Low operating and maintenance costs.
8. Small size and low weight.
9. Low noise and vibration; smooth operation.
10. Use of readily available materials and purchased components.
11. Prudent use of both uniquely designed parts and commercially available components.
12. Appearance that is attractive and appropriate to the application.

1-5 EXAMPLE OF THE INTEGRATION OF MACHINE ELEMENTS INTO A MECHANICAL DESIGN

Mechanical design is the process of designing and/or selecting mechanical components and putting them together to accomplish a desired function. Of course, machine elements must be compatible, must fit well together, and must perform safely and efficiently. The designer must consider not only the performance of the element being designed at a given time but also the elements with which it must interface.

To illustrate how the design of machine elements must be integrated with a larger mechanical design, let us consider the design of a speed reducer for the small tractor discussed in Section 1-4. Suppose that, to accomplish the speed reduction, you decide to design a double-reduction, spur gear reducer. You specify four gears, three shafts, six bearings, and a housing to hold the individual elements in proper relation to each other, as shown in the concept sketches in Figure 1-12.

The primary elements of the speed reducer in Figure 1-12 are:

1. The input shaft (shaft 1) is to be connected to the power source, a gasoline engine whose output shaft rotates at 2000 rpm. A flexible coupling is to be employed to minimize difficulties with alignment.
2. The first pair of gears, A and B, causes a reduction in the speed of the intermediate shaft (shaft 2) proportional to the ratio of the numbers of teeth in the gears. Gears B and C are both mounted to shaft 2 and rotate at the same speed.
3. A key is used at the interface between the hub of each gear and the shaft on which it is mounted to transmit torque between the gear and the shaft.
4. The second pair of gears, C and D, further reduces the speed of gear D and the output shaft (shaft 3) to the range of 290 to 295 rpm.
5. The output shaft is to carry a chain sprocket (not shown). The chain drive ultimately is to be connected to the drive wheels of the tractor.
6. Each of the three shafts is supported by two ball bearings, making them statically determinate and allowing the analysis of forces and stresses using standard principles of mechanics.
7. The bearings are held in a housing that is to be attached to the frame of the tractor. Note that the manner of holding each bearing is such that the inner race rotates with the shaft, while the outer race is held stationary.
8. Seals are shown on the input and output shafts to prohibit contaminants from entering the housing.
9. Other parts of the housing are shown schematically. Details of how the active elements are to be installed, lubricated, and aligned are only suggested at this stage of the design process to demonstrate feasibility. One possible assembly process could be as follows:
 - Start by placing the gears, keys, spacers, and bearings on their respective shafts.
 - Then insert shaft 1 into its bearing seat on the left side of the housing.
 - Insert the left end of shaft 2 into its bearing seat while engaging the teeth of gears A and B.
 - Install the center bearing support to provide support for the bearing at the right side of shaft 1.
 - Install shaft 3 by placing its left bearing into the seat on the center bearing support while engaging gears C and D.
 - Install the right side cover for the housing while placing the final two bearings in their seats.

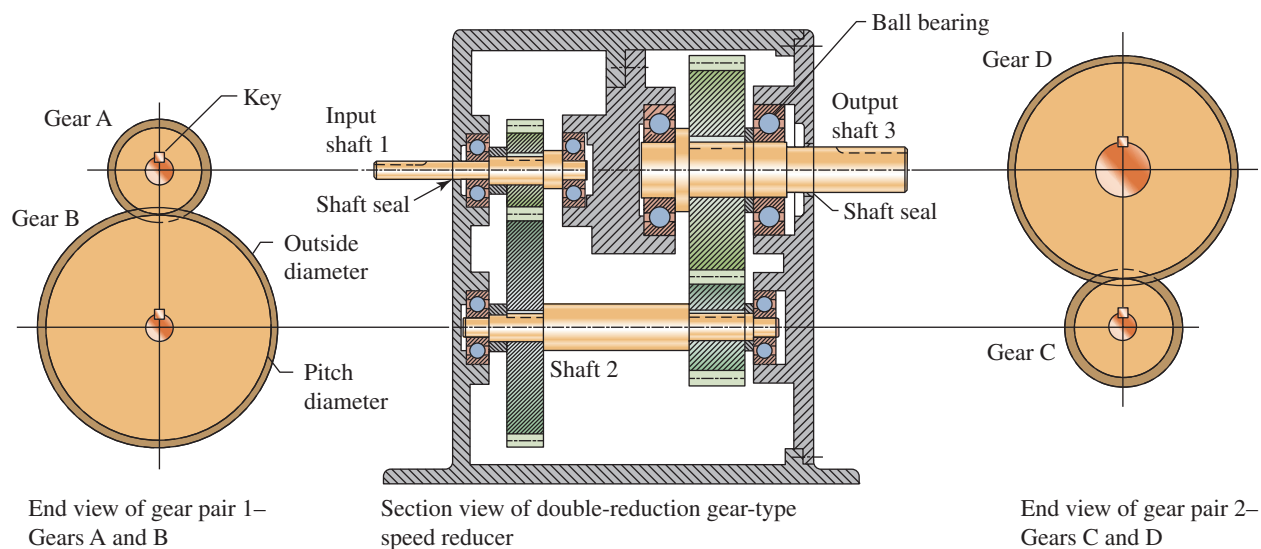


FIGURE 1-12 Conceptual design for a speed reducer

- Ensure careful alignment of the shafts.
- Place gear lubricant in the lower part of the housing.

Figures 9–29 to 9–31 show three examples of commercially available double-reduction gear reducers where you can see these details.

The arrangement of the gears, the placement of the bearings so that they straddle the gears, and the general configuration of the housing are also design decisions. The design process cannot rationally proceed until these kinds of decisions are made. Notice that the sketch of Figure 1–12 is where *integration* of the elements into a whole design begins. When the overall design is conceptualized, the design of the individual machine elements in the speed reducer can proceed. As each element is discussed, scan the relevant chapters of the book. Part II of this book, Chapters 7–15, provides details for design of the elements of the reducer. You should recognize that you have already made many design decisions by rendering such a sketch. First, you chose *spur gears* rather than helical gears, a worm and wormgear, or bevel gears. In fact, other types of speed-reduction devices—belt drives, chain drives, or many others—could be appropriate.

Gears

For the gear pairs, you must specify the number of teeth in each gear, the pitch (size) of the teeth, the pitch diameters, the face width, and the material and its heat treatment. These specifications depend on considerations of strength and wear of the gear teeth and the motion requirements (Chapters 8 and 9). You must also recognize that the gears must be mounted on shafts in a manner that ensures proper location of the gears, adequate torque transmitting capability from the gears to the shafts (as through keys), and safe shaft design.

Shafts

Having designed the gear pairs, you next consider the shaft design (Chapter 12). The shaft is loaded in bending and torsion because of the forces acting at the gear teeth. Thus, its design must consider strength and rigidity, and it must permit the mounting of the gears and bearings. Shafts of varying diameters may be used to provide shoulders against which to seat the gears and bearings. There may be keyseats cut into the shaft (Chapter 11). The input and output shafts will extend beyond the housing to permit coupling with the engine and the drive axle. The type of coupling must be considered, as it can have a dramatic effect on the shaft stress analysis (Chapter 11). Seals on the input and output shafts protect internal components (Chapter 11).

Bearings

Design of the bearings (Chapter 14) is next. If rolling contact bearings are to be used, you will probably select commercially available bearings from a manufacturer's

catalog, rather than design a unique one. You must first determine the magnitude of the loads on each bearing from the shaft analysis and the gear designs. The rotational speed and reasonable design life of the bearings and their compatibility with the shaft on which they are to be mounted must also be considered. For example, on the basis of the shaft analysis, you could specify the minimum allowable diameter at each bearing seat location to ensure safe stress levels. The bearing selected to support a particular part of the shaft, then, must have a bore (inside diameter) no smaller than the safe diameter of the shaft. Of course, the bearing should not be grossly larger than necessary. When a specific bearing is selected, the diameter of the shaft at the bearing seat location and allowable tolerances must be specified, according to the bearing manufacturer's recommendations, to achieve proper operation and life expectancy of the bearing.

Keys

Now the keys (Chapter 11) and the keyseats can be designed. The diameter of the shaft at the key determines the key's basic size (width and height). The torque that must be transmitted is used in strength calculations to specify key length and material. Once the working components are designed, the housing design can begin.

Housing

The housing design process must be both creative and practical. What provisions should be made to mount the bearings accurately and to transmit the bearing loads safely through the case to the structure on which the speed reducer is mounted? How will the various elements be assembled into the housing? How will the gears and bearings be lubricated? What housing material should be used? Should the housing be a casting, a weldment, or an assembly of machined parts?

The design process as outlined here implies that the design can progress in sequence: From the gears to the shafts, to the bearings, to the keys and couplings, and finally to the housing. It would be rare, however, to follow this logical path only once for a given design. Usually the designer must go back many times to adjust the design of certain components affected by changes in other components. This process, called *iteration*, continues until an acceptable overall design is achieved. Frequently prototypes are developed and tested during iteration.

Chapter 15 shows how all of the machine elements are finally integrated into a unit.

1-6 COMPUTATIONAL AIDS

Because of the usual need for several iterations and because many of the design procedures require long, complex calculations, spreadsheets, mathematical analysis software, computer programs, or programmable calculators are often useful in performing the design

analysis. Interactive spreadsheets or programs allow you, the designer, to make design decisions during the design process. In this way, many trials can be made in a short time, and the effects of changing various parameters can be investigated. Spreadsheets using Microsoft Excel are used most frequently as examples in this book for computer-aided design and analysis calculations.

Spreadsheets, and commercial software must be used carefully and it is recommended that the following statements guide your use of such aids:

- *Users of computer software and calculation aids must have a solid understanding of the relevant principles of design and stress analysis to ensure that design decisions are based on reliable foundations.*
- *Software should be used only after mastering a given design methodology by careful study and practicing manual techniques.*

Examples of commercially available mechanical design software are listed as Internet Sites 4–6 and 19. Internet sites for later chapters include several additional sources for software.

1–7 DESIGN CALCULATIONS

As you study this book and as you progress in your career as a designer, you will make many design calculations. It is important to record the calculations neatly, completely, and in an orderly fashion. You may have to explain to others how you approached the design, which data you used, and which assumptions and judgments you made. In some cases, someone else will actually check your work when you are not there to comment on it or to answer questions. Also, an accurate record of your design calculations is often useful if changes in design are likely. In all of these situations, you are going to be asked to communicate your design to someone else in written and graphic form.

To prepare a careful design record, you will usually take the following steps:

1. Identify the machine element being designed and the nature of the design calculation.
2. Draw a sketch of the element, showing all features that affect performance or stress analysis.
3. Show in a sketch the forces acting on the element (the free-body diagram), and provide other drawings to clarify the actual physical situation.
4. Identify the kind of analysis to be performed, such as stress due to bending, deflection of a beam, buckling of a column, and so on.
5. List all given data and assumptions.
6. Write the formulas to be used in symbol form, and clearly indicate the values and units of the variables involved. If a formula is not well known to a potential reader of your work, give the source. The reader may want to refer to it to evaluate the appropriateness of the formula.

7. Solve each formula for the desired variable.
8. Insert data, check units, and perform computations.
9. Judge the reasonableness of the result.
10. If the result is not reasonable, change the design decisions and recompute. Perhaps a different geometry or material would be more appropriate.
11. When a reasonable, satisfactory result has been achieved, specify the final values for all important design parameters, using standard sizes, convenient dimensions, readily available materials, and so on.

Figure 1–13 shows a sample design calculation. A beam is to be designed to span a 60-in pit to support a large gear weighing 2050 pounds (lb). The design assumes that a rectangular shape is to be used for the cross section of the beam. Other practical shapes could have been used. The objective is to compute the required dimensions of the cross section, considering both stress and deflection. A material for the beam is also chosen. Refer to Chapter 3 for a review of stress due to bending.

1–8 PREFERRED BASIC SIZES, SCREW THREADS, AND STANDARD SHAPES

One responsibility of a designer is to specify the final dimensions for load-carrying members. After completing the analyses for stress and deformation (strain), the designer will know the minimum acceptable values for dimensions that will ensure that the member will meet performance requirements. The designer then typically specifies the final dimensions to be standard or convenient values that will facilitate the purchase of materials and the manufacture of the parts. This section presents some guides to aid in these decisions and specifications.

Preferred Basic Sizes

Table A2–1 lists preferred basic sizes for fractional-inch, decimal-inch, and metric sizes.¹ You should choose one of these preferred sizes as the final part of your design. An example is at the end of the sample design calculation shown in Figure 1–13. You may, of course, specify another size if there is a sound functional reason.

American Standard Screw Threads

Threaded fasteners and machine elements having threaded connections are manufactured according to standard dimensions to ensure interchangeability of parts and to permit convenient manufacture with standard machines and tooling. Table A2–2 gives the dimensions of American

¹Throughout this book, some references to tables and figures have the letter A included in their numbers; these tables and figures are in the appendices in the back of the book. For example, Table A2–1 is the first table in Appendix 2; Figure A15–4 is the fourth figure in Appendix 15. These tables and figures are clearly identified in their captions in the appendices.

R. L. MOTT

DESIGN OF A BAR TO SUPPORT A GEAR IN A SOAKING PIT

BAR IS TO BE 60 IN LONG BETWEEN SUPPORTS
 GEAR WEIGHT 2050 LB
 HANGERS TO BE 24 IN APART

BAR IS A BEAM IN BENDING

$$\textcircled{1} \quad \sigma = M/S$$

ASSUME A RECTANGULAR SHAPE

S = SECTION MODULUS

$$S = th^2/6$$

LET $h \approx 3t$

$$\text{THEN } S = t(3t)^2/6 = 9t^3/6$$

$$S = 1.5 t^3$$

$$\textcircled{2} \text{ REQUIRED } t = \sqrt[3]{S/1.5}$$

TRY SAE 1040 HR STEEL BAR

$$S_y = 42000 \text{ PSI (YIELD STRENGTH)}$$

LET $\sigma = \sigma_d = S_y/N$ = DESIGN STRESS

N = DESIGN FACTOR

LET $N = 2$ (DEAD LOAD)

$$\sigma_d = 42000/2 = 21000 \text{ PSI}$$

THEN FROM $\textcircled{1}$: $S = M/\sigma_d$ = REQUIRED SECTION MODULUS

$$S = \frac{18450 \text{ LB}\cdot\text{IN}}{21000 \text{ LB/IN}^2} = 0.879 \text{ IN}^3$$

FROM $\textcircled{2}$

$$t = \sqrt[3]{S/1.5} = \sqrt[3]{0.879 \text{ IN}^3/1.5} = 0.837 \text{ IN}$$

$$\text{THEN } h = 3t = 3(0.837 \text{ IN}) = 2.51 \text{ IN}$$

SUPPLIER HAS $3/4 \times 2 3/4$ AVAILABLE

$$[h/t = 2.75/0.75 = 3.67 \text{ ok}]$$

$$\text{CHECK } S = th^2/6 = (0.75 \text{ IN})(2.75 \text{ IN})^2/6 = 0.945 \text{ IN}^3 > 0.879 \text{ IN}^3 \text{ ok}$$

$$\sigma = M/S = 18450 \text{ LB}\cdot\text{IN}/0.945 \text{ IN}^3 = 19500 \text{ PSI}$$

$$N = S_y/\sigma = 42000 \text{ PSI}/19500 \text{ PSI} = 2.15 \text{ ok}$$

$$\text{CHECK DEFLECTION AT CENTER: } y = \frac{Wa}{24 Ei} (3\ell^2 - 4a^2) \left(\frac{\text{REF}}{\text{Appendix A14-1(c)}} \right)$$

$$y = \frac{(1025)(18) [3(60)^2 - 4(18)^2]}{24 (30 \times 10^6) (1.30)} = 0.187 \text{ IN} \quad \text{ACCEPTABLE} \quad I = th^3/12 = \frac{(0.75)(2.75)^3}{12} = 1.30 \text{ IN}^4$$

SPECIFY: $3/4 \times 2 3/4$ RECTANGULAR STEEL BAR, SAE 1040 HR

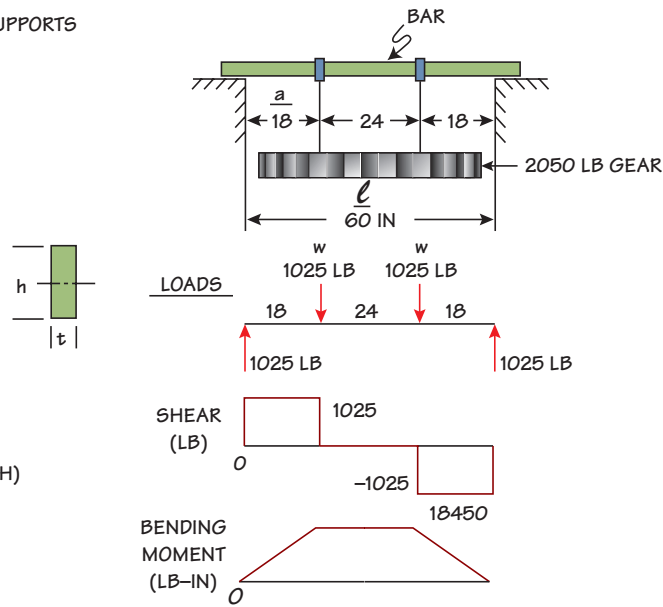


FIGURE 1-13 Sample design calculation

Standard Unified threads. Sizes smaller than 1/4 in are given numbers from 0 to 12, while fractional-inch sizes are specified for 1/4 in and larger sizes. Two series are listed: UNC is the designation for coarse threads, and UNF designates fine threads. Standard designations are as follows:

6-32 UNC	(number size 6, 32 threads per inch, coarse thread)
12-28 UNF	(number size 12, 28 threads per inch, fine thread)
$\frac{1}{2}$ -13 UNC	(fractional size 1/2 in, 13 threads per inch, coarse thread)
$1\frac{1}{2}$ -12 UNF	(fractional size $1\frac{1}{2}$ in, 12 threads per inch, fine thread)

The *pitch* of a screw thread, P , is the distance between corresponding points on two adjacent threads and it is the distance the screw would move axially

when the screw is turned one complete revolution. In the American Standard Unified thread system,

$$P = 1/n = 1/\text{number of threads per inch}$$

For example, for the 1/2-13 UNC thread, $P = 1/n = 1/13 = 0.0769$ in. For metric screw threads, described next, the pitch is included as part of the thread designation system. It is given as an axial distance between adjacent threads in mm.

Given in the tables are the basic major diameter (D), the number of threads per inch (n), and the tensile stress area (A_t), found from

⇒ Tensile Stress Area for Threads

$$A_t = 0.7854 \left(D - \frac{0.9743}{n} \right)^2 \quad (1-1)$$

When a threaded member is subjected to direct tension, the tensile stress area is used to compute the average tensile stress. It is based on a circular area computed from the mean of the pitch diameter and the minor diameter of the threaded member.

Metric Screw Threads

Table A2-3 gives similar dimensions for metric threads. Standard metric thread designations are of the form

$$M10 \times 1.5$$

where M stands for metric

The following number is the basic major diameter, D , in mm

The last number is the pitch, P , between adjacent threads in mm

The tensile stress area for metric threads is computed from the following equation and is based on a slightly different diameter. (See Reference 11, in the section, *Calculating Thread Tensile-Stress Area*.)

$$A_t = 0.7854(D - 0.9382P)^2 \quad (1-2)$$

Thus, the designation above would denote a metric thread with a basic major diameter of $D = 10.0$ mm and a pitch of $P = 1.5$ mm. Note that pitch = $1/n$. The tensile stress area for this thread is 58.0 mm^2 .

Commercially Available Shapes for Load-Carrying Members

An extensive array of shapes for the cross sections of load-carrying members is included in the 19 tables of Appendix 15. Many are available in either steel or aluminum.

Both U.S. Customary sizes and metric sizes are included with sizes ranging from quite small [about 10 mm or 3/8 in (0.375 in)] to large (up to 300 mm or 24 in). Note that the metric sizes listed are designed specifically in metric dimensions, rather than being *soft converted* from inch-sizes.

You are advised to refer first to the introduction that precedes the tables at the start of this appendix where the units, basic shape descriptions, available materials, and available sizes are given. This will guide you to appropriate tables for your application. In general, the following types of shapes are included:

- Tables A15-1 to A15-3: Angles or L-shapes
- Tables A15-4 to A15-8: Channels or C-shapes
- Tables A15-9 to A15-13: I-beam shapes, such as wide-flange (W) shapes, American Standard S-shapes, Aluminum Association I-beam shapes, European standard I-beam shapes
- Tables A15-14 to A15-16: Hollow square and rectangular tubing
- Tables A15-17 to A15-19: Pipe and hollow mechanical tubing

In many tables, links to Internet sites are given to allow searching for a much larger set of sizes for the types of shapes in the given table. You are encouraged to use this feature for problems and design exercises in this book.

Footnotes for many tables list the density, taken to be *mass density*, for steel and/or aluminum as an aid in computing the mass or weight of a given design. Note that:

Mass = Density $\times$ Volume, where

$$\text{Density} = \text{mass/unit volume} (\text{kg/m}^3 \text{ or } \text{lb}_m/\text{in}^3)$$

On earth, it is typically sufficiently accurate to assume that the U.S. Customary force unit, lb_f , is numerically equal to the mass unit, lb_m . For example, if density is given to be $0.283 \text{ lb}_m/\text{ft}^3$, it is reasonable to say that the weight density (also called specific weight) is $0.283 \text{ lb}_f/\text{ft}^3$. To obtain weight in the metric system, it is necessary to multiply the mass by g , the acceleration due to gravity. The nominal value of $g = 32.2 \text{ ft/s}^2$ or 9.81 m/s^2 . See Section 1-10 for additional discussion of the relationship between mass and weight.

Typical section properties listed for the shapes are:

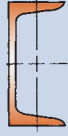
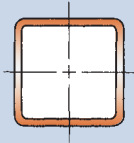
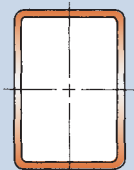
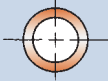
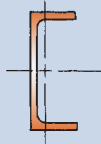
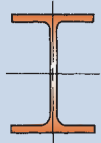
- A —Cross-sectional area
- I —Moment of inertia, sometimes called the second moment of the area; used for beam analysis and design
- S —Section modulus. Note that $S = I/c$, where c is the distance from the neutral axis to the outermost part of the section; used for beam analysis and design
- r —Radius of gyration. Note that $r = \sqrt{I/A}$; used for column analysis
- J —Polar moment of inertia; used for torsional analysis and design
- Z_p —Polar section modulus. Note that $Z_p = J/c$, where c is the outside radius of the pipe or tube; used for torsional analysis and design

Structural Shapes—Designations

Steel manufacturers provide a large array of standard structural shapes that are efficient in the use of material and that are convenient for specification and installation into building structures or machine frames. Included, as shown in Table 1-1, are standard angles (L-shapes), channels (C-shapes), wide-flange beams (W-shapes), American Standard beams (S-shapes), structural tubing, and pipe. Note that the W-shapes and the S-shapes are often referred to in general conversation as “I-beams” because the shape of the cross section looks like the capital letter I. See Reference 2.

Materials used for structural shapes are typically called *structural steels*, and their characteristics and properties are described more fully in Chapter 2. Refer to Appendix 7 for typical strength data. Rolled W-shapes are most readily available in ASTM A992, A572 Grade 50, or A36. S-shapes and C-shapes are typically made

TABLE 1-1 Designations for Steel and Aluminum Shapes

Name of shape	Shape	Symbol	Example designation and Appendix table
Angle		L	L4×3× $\frac{1}{2}$ Table A15-1
Channel		C	C15×50 Table A15-4
Wide-flange beam		W	W14×43 Table A15-9
American Standard beam		S	S10×35 Table A15-10
Structural tubing—square		HSS	4×4× $\frac{1}{4}$ Table A15-14
Structural tubing—rectangular		HSS	6×4× $\frac{1}{4}$ Table A15-14
Pipe		PIPE	4-inch standard weight 4-inch Schedule 40 Table A15-17
Aluminum Association channel		C	C4×1.738 Table A15-6
Aluminum Association I-beam		I	I 8×6.181 Table A15-11

from ASTM A572 Grade 50 or A36. ASTM A36 should be specified for steel angles and plates. Hollow structural shapes (HSS) are most readily available in ASTM A500.

Aluminum structural shapes are most often made from extruded 6061-T6 alloy.

Angles (L-Shapes)

Table A15–1 shows sketches of the typical shapes of steel angles having equal or unequal leg lengths. Called *L-shapes* because of the appearance of the cross section, angles are often used as tension members of trusses and towers, framing members for machine structures, lintels over windows and doors in construction, stiffeners for large plates used in housings and beams, brackets, and ledge-type supports for equipment. Some refer to these shapes as “angle iron.” The U.S. standard designation takes the following form, using one example size:

$$L4 \times 3 \times \frac{1}{2}$$

where L refers to the L-shape

4 is the length of the longer leg

3 is the length of the shorter leg

$\frac{1}{2}$ is the thickness of the legs

Dimensions are in inches

Smaller angles in U.S. units are listed in Table A15–2. Angles made to purely metric dimensions are listed in Table A15–3 over a range of sizes from 10 mm to 100 mm. Many of the listed shapes are available in either aluminum or steel.

Channels (C-Shapes)

See Table A15–4 for the appearance of American Standard channels and their geometric properties. Channels are used in applications similar to those described for angles. The flat web and the two flanges provide a generally stiffer shape than angles.

The sketch at the top of the table shows that channels have tapered flanges and webs with constant thickness. The slope of the flange taper is approximately 2 inches in 12 inches, and this makes it difficult to attach other members to the flanges. Special tapered washers are available to facilitate fastening. Note the designation of the x - and y -axes in the sketch, defined with the web of the channel vertical which gives it the characteristic C-shape. This is most important when using channels as beams or columns. The x -axis is located on the horizontal axis of symmetry, while the dimension x , given in the table, locates the y -axis relative to the back of the web. The centroid is at the intersection of the x - and y -axes.

The form of the U.S. standard designation for channels is

$$C15 \times 50$$

where C indicates that it is a standard C-shape

15 is the nominal (and actual) depth in inches with the web vertical

50 is the weight per unit length in lb/ft

Most other channels have flanges with uniform thicknesses that are often made by rolling flat sheets into the C-shape or by extrusion. Smaller channels in U.S. units are listed in Table A15–5. Aluminum angles made to European standards in purely metric dimensions are listed in Table A15–7, and steel metric channels are listed in Table A15–8. The designations for metric channels and small channels in U.S. units contain only size data, not weight per unit length.

Channels made to Aluminum Association standard shapes in U.S. units are listed in Table A15–6. Several designation systems are in common use. Some give the web height, leg length, and thickness. In this book, we use a form similar to that for standard steel shapes. For example,

$$C4 \times 1.738$$

where C indicates the basic shape

4 indicates the depth of the shape (web height) in inches

1.738 indicates the weight per unit length in lb/ft

I-Beam Shapes

Wide-Flange Shapes (W-Shapes). Refer to Table A15–9, which illustrates the most common shape used for beams. W-shapes have relatively thin webs and somewhat thicker, flat flanges with constant thickness. Most of the area of the cross section is in the flanges, farthest away from the horizontal centroidal axis (x -axis), thus making the moment of inertia very high for a given amount of material. Note that the properties of moment of inertia and section modulus are very much higher with respect to the x -axis than they are for the y -axis. Therefore, W-shapes are typically used in the orientation shown in the sketch in Table A15–9. Also, these shapes are best when used in pure bending without twisting because they are quite flexible in torsion.

The standard designation for steel W-shapes carries much information. Consider the following example:

$$W14 \times 43$$

where W indicates that it is a W-shape

14 is the nominal depth in inches

43 is the weight per unit length in lb/ft

The term *depth* is the standard designation for the vertical height of the cross section when placed in the orientation shown in Table A15–9. Note from the data in the table that the actual depth is often different from

the nominal depth. For the W14×43, the actual depth is 13.66 in.

All sizes listed in Table A15–9, along with many more (see Reference 2), are available in steel. A few of the listed sizes are also available in aluminum.

American Standard Beams (S-Shapes). Table A15–10 shows the properties for S-shapes. Much of the discussion given for W-shapes applies to S-shapes as well. Note that, again, the weight per foot of length is included in the designation such as the S10×35, which weighs 35 lb/ft. For most, but not all, of the S-shapes, the actual depth is the same as the nominal depth. The flanges of the S-shapes are tapered at a slope of approximately 2 inches in 12 inches, similar to the flanges of the C-shapes. The x - and y -axes are defined as shown with the web vertical.

Often wide-flange shapes (W-shapes) are preferred over S-shapes because of their relatively wide flanges, the constant thickness of the flanges, and the generally higher section properties for a given weight and depth.

A few of the sizes for S-shapes are available in aluminum.

Aluminum Association I-Beam Shapes. Aluminum Association standard I-beam shapes in U.S. units are listed in Table A15–11. Several designation systems are in common use. Some give the web height, flange width, and the thickness for either the web or the flange. In this book, we use a form similar to that for standard steel I-shapes. For example,

$$I\ 8\times 6.181$$

where I indicates the basic I-shape

8 indicates the depth of the shape (web height) in inches

6.181 indicates the weight per unit length in lb/ft

Other I-Beam Shapes. Table A15–12 lists small extruded aluminum shapes. European standard steel shapes in purely metric units are listed in Table A15–13.

Hollow Tubing (Square and Rectangular)

Square tubing and rectangular tubing are very useful in machine structures because they provide good section properties for members loaded as beams in bending and for torsional loading (twisting) because of the closed cross section. The flat sides often facilitate fastening of members together or the attachment of equipment to the structural members. Some frames are welded into an integral unit that functions as a stiff space-frame. Square tubing makes an efficient section for columns.

See Table A15–14 for the appearance and properties for hollow steel structural shapes. These shapes, often called HSS, are usually formed from flat sheet and

welded along the length. The section properties account for the corner radii. Note the sketches showing the x - and y -axes. The standard designation takes the form

$$\text{HSS } 6\times 4\times \frac{1}{4}$$

where 6 is the depth of the longer side in inches

4 is the width of the shorter side in inches

$\frac{1}{4}$ is the wall thickness in inches

Note that for standard HSS shapes as listed in Reference 2, the *design wall thickness*, t_w , from Table A15–14 should be used. That value, *smaller* than the nominal size used in the designation, is used to compute the listed section properties.

Table A15–15 lists smaller steel and aluminum square and rectangular tubing in inch-sizes from 0.375 in (3/8 in) to 3.00 in depth. Metric sizes for square and rectangular tubing are listed in Table A15–16 from 20 mm to 300 mm depth.

Pipe and Hollow Circular Tubing

Hollow circular sections, commonly called *pipe*, are very efficient for use as beams, torsion members, and columns. The placement of the material uniformly away from the center of the pipe enhances the moment of inertia for a given amount of material and gives the pipe uniform properties with respect to all axes through the center of the cross section. The closed cross-sectional shape gives it high strength and stiffness in torsion as well as in bending.

Table A15–17 gives the properties for American National Standard Schedule 40 welded and seamless wrought steel pipe. This type of pipe is often used to transport water and other fluids, but it also performs well in structural applications. Note that the actual inside and outside diameters are somewhat different from the nominal size, except for the very large sizes. Construction pipe is often called *Standard Weight Pipe*, and it has the same dimensions as the Schedule 40 pipe for sizes from 1/2 in to 10 in. Other “schedules” and “weights” of pipe are available with larger and smaller wall thicknesses.

Other hollow circular sections are commonly available that are referred to as *tubing*. These sections are available in carbon steel, alloy steel, stainless steel, aluminum, copper, brass, titanium, and other materials. See References 1, 2, 5, and 9 for a variety of types and sizes of pipe and tubing.

Tubing is typically specified by its outside diameter and wall thickness, with the wall thickness sometimes given as a standard *gauge*. Table A15–18 lists tubing in U.S. sizes in steel and aluminum with outside diameters from 0.50 in to 5.0 in and various wall thicknesses. Table A15–19 lists tubing in metric sizes in steel and aluminum with outside diameters from 10 mm to 150 mm in various wall thicknesses.

1-9 UNIT SYSTEMS

We will perform computations in this book by using either the U.S. Customary Unit System (inch-pound-second) or the International System (SI). Table 1-2 lists the typical units used in the study of machine design. *SI*, the abbreviation for “Le Système International d’Unités,” is the standard for metric units throughout the world. (See Reference 4.) For convenience, the term *SI units* will be used instead of *metric units*.

Prefixes applied to the basic units indicate order of magnitude. Only those prefixes listed in Table 1-3, which differ by a factor of 1000, should be used in technical

calculations. The final result for a quantity should be reported as a number between 0.1 and 10 000, times some multiple of 1000. Then the unit with the appropriate prefix should be specified. Table 1-4 lists examples of proper SI notation.

Sometimes you have to convert a unit from one system to another. Appendix 16 provides tables of conversion factors. Also, you should be familiar with the typical order of magnitude of the quantities encountered in machine design so that you can judge the reasonableness of design calculations (see Table 1-5 for several examples).

TABLE 1-2 Typical Units used in Machine Design

Quantity	U.S. Customary unit	SI unit
Length or distance	inch (in) foot (ft)	meter (m) millimeter (mm)
Area	square inch (in ²)	square meter (m ²) or square millimeter (mm ²)
Force	pound (lb) kip (K) (1000 lb)	newton (N) (1 N = 1 kg · m/s ²)
Mass	slug (lb · s ² /ft)	kilogram (kg)
Time	second (s)	second (s)
Angle	degree (°)	radian (rad) or degree (°)
Temperature	degrees Fahrenheit (°F)	degrees Celsius (°C)
Torque or moment	pound-inch (lb · in) or pound-foot (lb · ft)	newton-meter (N · m)
Energy or work	pound-inch (lb · in)	joule (J) 1 J = 1 N · m
Power	horsepower (hp) (1 hp = 550 lb · ft/s)	watt (W) or kilowatts (kW) (1 W = 1 J/s = 1 N · m/s)
Stress, pressure, or modulus of elasticity	pounds per square inch (lb/in ² , or psi) kips per square inch (K/in ² , or ksi)	pascal (Pa) (1 Pa = 1 N/m ²) kilopascal (kPa) (1 kPa = 10 ³ Pa) megapascal (MPa) (1 MPa = 10 ⁶ Pa) gigapascal (GPa) (1 GPa = 10 ⁹ Pa)
Section modulus	inches cubed (in ³)	meters cubed (m ³) or millimeters cubed (mm ³)
Moment of inertia	inches to the fourth power (in ⁴)	meters to the fourth power (m ⁴) or millimeters to the fourth power (mm ⁴)
Rotational speed	revolutions per min (rpm)	radians per second (rad/s)

TABLE 1-3 Prefixes used with SI Units

Prefix	SI symbol	Factor
micro-	μ	$10^{-6} = 0.000\,001$
milli-	m	$10^{-3} = 0.001$
kilo-	k	$10^3 = 1000$
mega-	M	$10^6 = 1\,000\,000$
giga-	G	$10^9 = 1\,000\,000\,000$

TABLE 1-4 Quantities Expressed in SI Units

Computed result	Reported result
0.001 65 m	1.65×10^{-3} m, or 1.65 mm
32 540 N	32.54×10^3 N, or 32.54 kN
1.583×10^5 W	158.3×10^3 W, or 158.3 kW; or 0.1583×10^6 W; or 0.1583 MW
2.07×10^{11} Pa	207×10^9 Pa, or 207 GPa

TABLE 1-5 Typical Order of Magnitude for Commonly Encountered Quantities

Quantity	U.S. Customary unit	SI unit
Dimensions of a wood standard 2 × 4	1.50 in × 3.50 in	38 mm × 89 mm
Moment of inertia of a 2 × 4 (3.50-in side vertical)	5.36 in ⁴	2.23 × 10 ⁶ mm ⁴ , or 2.23 × 10 ⁻⁶ m ⁴
Section modulus of a 2 × 4 (3.50-in side vertical)	3.06 in ³	5.02 × 10 ⁴ mm ³ , or 5.02 × 10 ⁻⁵ m ³
Force required to lift 1.0 gal of gasoline	6.01 lb	26.7 N
Density of water	1.94 slugs/ft ³	1000 kg/m ³ , or 1.0 Mg/m ³
Compressed air pressure in a factory	100 psi	690 kPa
Yield point of SAE 1040 hot-rolled steel	42 000 psi, or 42 ksi	290 MPa
Modulus of elasticity of steel	30 000 000 psi, or 30 × 10 ⁶ psi	207 GPa

Example Problem 1-1

Express the diameter of a shaft in millimeters if it is measured to be 2.755 in.

Solution

Table A16 gives the conversion factor for length to be 1.00 in = 25.4 mm. Then

$$\text{Diameter} = 2.755 \text{ in} \frac{25.4 \text{ mm}}{1.00 \text{ in}} = 69.98 \text{ mm}$$

Example Problem 1-2

An electric motor is rotating at 1750 revolutions per minute (rpm). Express the speed in radians per second (rad/s).

Solution

A series of conversions is required.

$$\text{Rotational speed} = \frac{1750 \text{ rev}}{\text{min}} \frac{2\pi \text{ rad}}{\text{rev}} \frac{1 \text{ min}}{60 \text{ s}} = 183.3 \text{ rad/s}$$

1-10 DISTINCTION AMONG WEIGHT, FORCE, AND MASS

Distinction must be made among the terms *force*, *mass*, and *weight*. *Mass* is the quantity of matter in a body. A *force* is a push or pull applied to a body that results in a change in the body's motion or in some deformation of the body. Clearly these are two different physical phenomena, but the distinction is not always understood. The units for force and mass used in this text are listed in Table 1-2.

The term *weight*, as used in this book, refers to the amount of *force* required to support a body against the influence of gravity. Thus, in response to “What is the weight of 75 kg of steel?” we would use the relationship between force and mass from physics:

Weight/Mass Relationship

$$F = ma \quad \text{or} \quad w = mg$$

where F = force

m = mass

a = acceleration

w = weight

g = acceleration due to gravity

We will use

$$g = 32.2 \text{ ft/s}^2 \quad \text{or} \quad g = 9.81 \text{ m/s}^2$$

Then, to compute the weight,

$$w = mg = 75 \text{ kg} (9.81 \text{ m/s}^2)$$

$$w = 736 \text{ kg} \cdot \text{m/s}^2 = 736 \text{ N}$$

Remember that, as shown in Table 1-2, the newton (N) is equivalent to 1.0 kg · m/s². In fact, the newton is defined as the force required to give a mass of 1.0 kg an acceleration of 1.0 m/s². In our example, then, we would say that the 75-kg mass of steel has a weight of 736 N.

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INTERNET SITES FOR GENERAL MECHANICAL DESIGN

Included here are Internet sites that can be used in many of the chapters of this book and in general design practice to identify commercial suppliers of machine elements and standards for design or to perform stress analyses. Later chapters include sites specific to the topics covered there.

1. **American National Standards Institute (ANSI)** A private, nonprofit organization that administers and coordinates the U.S. voluntary standardization and conformity assessment system.
2. **Global Engineering Documents** A database of standards and publications offered by many standards-developing organizations such as ASME, ASTM, SAE, DIN, ISO and several other organizations.
3. **GlobalSpec** A database of a wide variety of technical products and services that provides for searching by technical specifications, access to supplier information, and comparison of suppliers for a given product. The Mechanical Components category includes many of the topics addressed in this book.
4. **MDSOLIDS** Educational software for strength of materials topics, including beams, flexure, torsion members, columns, axial structures, statically indeterminate structures, trusses, section properties, and Mohr's circle analysis. This software may serve as a review tool for the prerequisite knowledge needed in this book.
5. **Software for Structures - engineering-software.com** Source for variety of software products including beam analysis and trusses.

6. **SkyCiv Online Engineering Software.** Producer of the SkyCiv brand of software for beam analysis, trusses, frames, shafts and other applications.
7. **Power Transmission Home Page** Clearinghouse on the Internet for buyers, users, and sellers of power transmission products and services. Included are gears, gear drives, belt drives, chain drives, bearings, clutches, brakes, and many other machine elements covered in this book.

INTERNET SITES FOR INNOVATION AND MANAGING COMPLEX DESIGN PROJECTS

8. **Axiomatic Design Solutions, Inc.** An integrated method of combining customer needs and functional requirements to produce detailed design parameters, the final embodiment of the design solution, and the process needed to produce the design efficiently.
9. **QFD Institute** An organization promoting the Quality Function Deployment (QFD) method for product design and quality systems management.
10. **Applied Marketing Science (AMS), Inc.** Consulting firm providing voice of the customer (VOC) insights and using Quality Function Deployment (QFD) based on the “House of Quality” matrix described in Reference 8.
11. **iSixSigma.com** The mission of iSixSigma.com is to provide a free information resource to help business professionals successfully implement Lean Six Sigma and business process improvement tools and methodologies within their organizations.
12. **The TRIZ Journal.** A site offering case studies, theoretical articles, TRIZ history, symposia, classes, and books to help designers learn TRIZ.
13. **Technical Innovation Center, Inc.** Provider of TRIZ consulting, training, and publishing services.
14. **The Altshuller Institute for TRIZ Studies** Provider of services to the world TRIZ community, authorized by Genrikh Altshuller, the founder of TRIZ.
15. **Triz Consulting, Inc.** Provider of TRIZ training for systematic innovation, consulting, and publications.
16. **Systems2win.com** Provider of Excel templates for Lean Six Sigma to Define, Measure, Analyze, Improve, and Control quality (DMAIC 6 Σ tools).
17. **FMEA-FMECA.com** An Internet source of information about the Failure Modes and Effects Analysis (FMEA) and Failure Modes and Effects with Criticality Analysis (FMECA). Source for books, software, forms, examples, and other resources.
18. **Kepner-Tregoe, Inc.** Kepner-Tregoe provides consulting and training services to help clients implement their strategies by embedding problem-solving, decision-making, and project execution methods. Publisher of the book cited in Reference 25.
19. **TEDATA GMBH** Producer of the MDESIGN software, a large package of modules that address numerous mechanical design tasks such as gears, bearings, belt drives, chain drives, shafts, axles, beams, bolts, welded connections, springs, clutches, and others.

PROBLEMS

Functions and Design Requirements

For the devices described in Problems 1–14, write a set of functions and design requirements in a similar manner to those in Section 1–4. You or your instructor may add more specific information to the general descriptions given.

1. The hood latch for an automobile.
2. A hydraulic jack used for car repair.
3. A portable crane to be used in small garages and homes.
4. A machine to crush soft-drink or beer cans.
5. An automatic transfer device for a production line.
6. A device to raise a 55-gallon (gal) drum of bulk materials and dump the contents into a hopper.
7. A paper feed device for a copier.
8. A conveyor to elevate and load gravel into a truck.
9. A crane to lift building materials from the ground to the top of a building during construction.
10. A machine to insert toothpaste tubes into cartons.
11. A machine to insert 24 cartons of toothpaste into a shipping container.
12. A gripper for a robot to grasp a spare tire assembly and insert it into the trunk of an automobile on an assembly line.
13. A table for positioning a weldment in relation to a robotic welder.
14. A garage door opener.

Units and Conversions

For Problems 15–28, perform the indicated conversion of units. (Refer to Appendix 16 for conversion factors.) Express the results with the appropriate prefix as illustrated in Tables 1–3 and 1–4.

15. Convert a shaft diameter of 1.75 in to mm.
16. Convert the length of a conveyor from 46 ft to meters.
17. Convert the torque developed by a motor of 12 550 lb·in to N·m.
18. A wide-flange steel-beam shape, W12×14, has a cross-sectional area of 4.12 in². Convert the area to mm².
19. The W12×14 beam shape has a section modulus of 14.8 in³. Convert it to mm³.
20. The W12×14 beam shape has a moment of inertia of 88.0 in⁴. Convert it to mm⁴.
21. What standard steel equal leg angle would have a cross-sectional area closest to (but greater than) 750 mm²? See Appendix 15.
22. An electric motor is rated at 7.5 hp. What is its rating in watts (W)?
23. A vendor lists the ultimate tensile strength of a steel to be 127 000 psi. Compute the strength in MPa.
24. Compute the weight of a steel shaft, 35.0 mm in diameter and 675 mm long. (See Appendix 3 for the density of steel.)
25. A torsional spring requires a torque of 180 lb·in to rotate it 35°. Convert the torque to N·m and the rotation to radians. If the *scale of the spring* is defined as the applied torque per unit of angular rotation, compute the spring scale in both unit systems.

26. To compute the energy used by a motor, multiply the power that it draws by the time of operation. Consider a motor that draws 12.5 hp for 16 h/day, five days per week. Compute the energy used by the motor for one year. Express the result in $\text{ft} \cdot \text{lb}$ and $\text{W} \cdot \text{h}$.
27. One unit used for fluid viscosity in Chapter 16 of this book is the *reyn*, defined as $1.0 \text{ lb} \cdot \text{s}/\text{in}^2$. If a lubricating oil has a viscosity of 3.75 reyn, convert the viscosity to the standard units in the U.S. Customary System $\text{lb} \cdot \text{s}/\text{ft}^2$ and in the SI ($\text{N} \cdot \text{s}/\text{m}^2$).
28. The life of a bearing supporting a rotating shaft is expressed in number of revolutions. Compute the life of a bearing that rotates 1750 rpm continuously for 24 h/day for five years.

MATERIALS IN MECHANICAL DESIGN

The Big Picture

You Are the Designer

- 2-1 Objectives of This Chapter
- 2-2 Properties of Materials
- 2-3 Classification of Metals and Alloys
- 2-4 Variability of Material Properties Data
- 2-5 Carbon and Alloy Steel
- 2-6 Conditions for Steels and Heat Treatment
- 2-7 Stainless Steels
- 2-8 Structural Steel
- 2-9 Tool Steels
- 2-10 Cast Iron
- 2-11 Powdered Metals
- 2-12 Aluminum
- 2-13 Zinc Alloys and Magnesium
- 2-14 Nickel-based Alloys and Titanium
- 2-15 Copper, Brass, and Bronze
- 2-16 Plastics
- 2-17 Composite Materials
- 2-18 Materials Selection

THE BIG PICTURE

Materials in Mechanical Design

Discussion Map

- You must understand the behavior of materials to make good design decisions and to communicate with suppliers and manufacturing staff.

Discover

Examine consumer products, industrial machinery, automobiles, and construction machinery.

What materials are used for the various parts?

Why do you think those materials were specified?

How were they processed?

What material properties were important to the decisions to use particular materials?

Examine the appendices tables, and refer to them later as you read about specific materials.

This chapter summarizes the design properties of a variety of materials. The appendices include data for many examples of these materials in many conditions.

It is the designer's responsibility to specify suitable materials for each component of a mechanical device. Your initial efforts in specifying a material for a particular component of a mechanical design should be

directed to the basic kind of material to be used. Keep an open mind until you have specified the functions of the component, the kinds and magnitudes of loads it will carry, and the environment in which it must

operate. Your selection of a material must consider its physical and mechanical properties and match them to the expectations placed on it. First consider the following classes of materials:

Metals and their alloys	Plastics	Composites
Elastomers	Woods	Ceramics and glasses

Each of these classes contains a large number of specific materials covering a wide range of actual properties. However, you probably know from your experience the general behavior of each kind and have some feel for the applications in which each is typically used. Most of the applications considered in the study of design of machine elements in this book use metal alloys, plastics, and composites.

Satisfactory performance of machine components and systems depends greatly on the materials that the designer specifies. As a designer, you must understand how materials behave, what properties of the material affect the performance of the parts, and how you should interpret the large amounts of data available on material properties. Your ability to effectively communicate your specifications for materials with suppliers, purchasing agents, metallurgists, manufacturing process personnel, heat treatment personnel, plastics molders, machinists, and quality assurance specialists often has a strong influence on the success of a design.

Explore what kinds of materials are used in consumer products, industrial machinery, automobiles, construction machinery, and other devices and systems that you come into contact with each day. Make judgments about why each material was specified for a particular application. Where do you see steel being used? Contrast that usage with where aluminum or other nonferrous materials are used. How are the products produced? Can you find different parts that are machined, cast, forged, roll-formed, and welded? Why do you think those processes were specified for those particular products?

Document several applications for plastics and describe the different forms that are available and that have been made by different manufacturing processes. Which are made by plastic molding processes, vacuum forming, blow molding, and others? Can you identify parts made from composite materials that have a significant amount of high-strength fibers embedded in a plastic matrix? Check out sporting goods and parts of cars, trucks, and airplanes.

From the products that you found from the exploration outlined previously, identify the basic properties of the materials that were important to the designers: strength, rigidity (stiffness), weight (density), corrosion resistance, appearance, machinability, weldability, ease of forming, cost, and others.

This chapter focuses on material selection and the use of material property data in design decisions, rather than on the metallurgy or chemistry of the materials. One of the uses of the information in this chapter is as a glossary of terms that you can use throughout the book; important terms are given in *italic* type. Also, there are numerous references to Appendix 3 through 13, where tables of data for material properties are given. Go there now and see what kinds of data are provided. Then you can study the tables in more depth as you read the text. Note that many of the problems that you will solve in this book and the design projects that you complete will use data from these tables.

The subject of this chapter is very broad and it is not possible to include in this book all the detail you may need for each design situation that arises as you study this book or in your career. Huge amounts of additional information are available on the Internet and the numerous sites listed at the end of this chapter give suggestions that may lead to what you need. Many of the listed sites are referred to within this chapter, but some are not and they provide alternate ways for you to search for what you need.

Now apply some of what you have gained from **The Big Picture** exploration to a specific design situation as outlined in **You Are the Designer**, which follows.

YOU ARE THE DESIGNER

You are part of a team responsible for the design of an electric lawn mower for the household market. One of your tasks is to specify suitable materials for the various components. Consider your own experience with such lawn mowers and think what materials would be used for these key components: *wheels*, *axles*, *housing*, and *blade*. What are their functions? What conditions of service will each encounter? What is one reasonable type of material for each component and what general properties should it have? How could they be manufactured? Possible answers to these questions follow.

Wheels

Function: Support the weight of the mower. Permit easy, rolling movement. Provide for mounting on an axle. Ensure safe operation on flat or sloped lawn surfaces.

Conditions of service: Must operate on grass, hard surfaces, and soft earth. Exposed to water, lawn fertilizers, and general outdoor conditions. Will carry moderate loads. Requires an attractive appearance.

One reasonable material: One-piece plastic wheel incorporating the tire, rim, and hub. Must have good strength, stiffness, toughness, and wear resistance.

Manufacturing method: Plastic injection molding

Axles

Function: Transfer the weight of mower from the housing to the wheels. Allow rotation of the wheels. Maintain location of the wheels relative to the housing.

Conditions of service: Exposure to general outdoor conditions. Moderate loads.

One possible material: Steel rod with provisions for mounting wheels and attaching to housing. Requires moderate strength, stiffness, and corrosion resistance.

Manufacturing method: Commercially available cylindrical rod. Parts of the rod may need machining.

Housing

Function: Support, safely enclose, and protect operating components, including the blade and motor. Accommodate the attachment of two axles and a handle. Permit cut grass to exit the cutting area.

Conditions of service: Moderate loads and vibration due to motor. Possible shock loads from wheels. Multiple attachment points for axles, handle, and motor. Exposed to wet grass and general outdoor conditions. Requires attractive appearance.

One possible material: Heavy-duty plastic with good strength, stiffness, impact resistance, toughness, and weather resistance.

Manufacturing method: Plastic injection molding. May require machining for holes and mounting points for the motor.

Blade

Function: Cut blades of grass and weeds while rotating at high speed. Facilitate connection to motor shaft. Operate safely when foreign objects are encountered, such as stones, sticks, or metal pieces.

Conditions of service: Normally moderate loads. Occasional shock and impact loads. Must be capable of sharpening a portion of the blade to ensure clean cutting of grass. Maintain sharpness for reasonable time during use.

One possible material: Steel with high strength, stiffness, impact resistance, toughness, and corrosion resistance.

Manufacturing method: Stamping from flat steel strip. Machining and/or grinding for cutting edge.

This simplified example of the material selection process should help you to understand the importance of the information provided in this chapter about the behavior of materials commonly used in the design of machine elements. A more comprehensive discussion of material selection occurs at the end of this chapter.

2-1 OBJECTIVES OF THIS CHAPTER

After completing this chapter, you will be able to:

1. State the types of material properties that are important to the design of mechanical devices and systems.
2. Define the following terms: *tensile strength*, *yield strength*, *proportional limit*, *elastic limit*, *modulus of elasticity in tension*, *ductility* and *percent elongation*, *shear strength*, *Poisson's ratio*, *modulus of elasticity in shear*, *hardness*, *machinability*, *impact strength*, *creep density*, *coefficient of thermal expansion*, *thermal conductivity*, and *electrical resistivity*.
3. Describe the nature of *carbon and alloy steels*, the number-designation system for steels, and the effect of several kinds of alloying elements on the properties of steels.
4. Describe the manner of designating the condition and heat treatment of steels, including *hot rolling*, *cold drawing*, *annealing*, *normalizing*, *through-hardening*, *tempering*, and *case hardening by flame hardening*, *induction hardening*, and *carburizing*.
5. Describe *stainless steels* and recognize many of the types that are commercially available.
6. Describe *structural steels* and recognize many of their designations and uses.

7. Describe *cast irons* and several kinds of *gray iron*, *ductile iron*, and *malleable iron*.
8. Describe *powdered metals* and their properties and uses.
9. Describe several types of *tool steels* and *carbides* and their typical uses.
10. Describe *aluminum alloys* and their conditions, such as *strain hardening* and *heat treatment*.
11. Describe the nature and typical properties of *zinc*, *titanium*, *copper*, *brass*, *bronze*, and *nickel-based alloys*.
12. Describe several types of *plastics*, both *thermosetting* and *thermoplastic*, and their typical properties and uses.
13. Describe several kinds of *composite materials* and their typical properties and uses.
14. Implement a rational material selection process.

2-2 PROPERTIES OF MATERIALS

Machine elements are very often made from one of the metals or metal alloys such as steel, aluminum, cast iron, zinc, titanium, or bronze. This section describes the important properties of materials as they affect mechanical design.

Strength, elastic, and ductility properties for metals, plastics, and other types of materials are usually determined from a *tensile test* in which a sample of the material, typically in the form of a round or flat bar, is clamped between jaws and pulled slowly until it breaks in tension. The magnitude of the force on the bar and the corresponding change in length (strain) are monitored and recorded continuously during the test. Because the

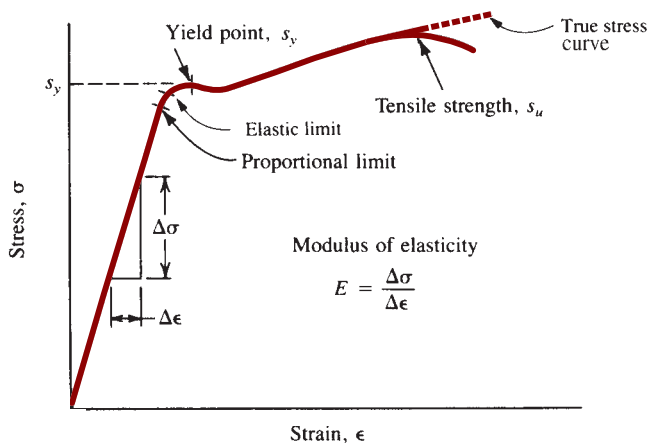


FIGURE 2-1 Typical stress-strain diagram for steel

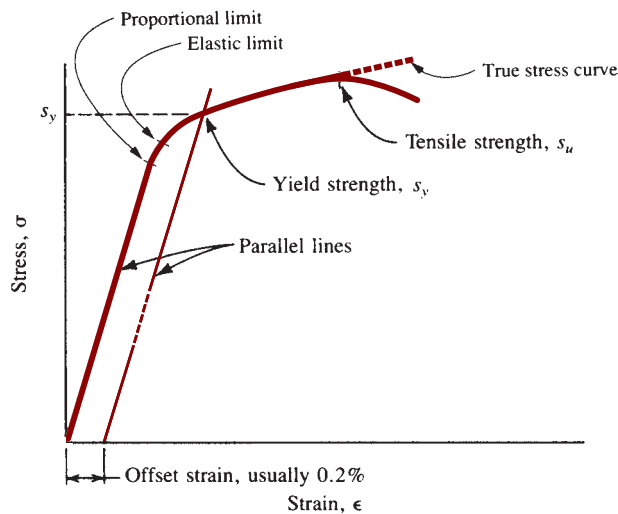


FIGURE 2-2 Typical stress-strain diagram for aluminum and other metals having no yield point

stress in the bar is equal to the applied force divided by the area, stress is proportional to the applied force. The data from such tensile tests are often shown on *stress-strain diagrams*, such as those shown in Figures 2-1 and 2-2. In the following paragraphs, several strength, elastic, and ductility properties of metals are defined.

Tensile Strength, s_u

The peak of the stress-strain curve is considered the *ultimate tensile strength* (s_u), sometimes called the *ultimate strength* or simply the *tensile strength*. At this point during the test, the highest *apparent stress* on a test bar of the material is measured. As shown in Figures 2-1 and 2-2, the curve appears to drop off after the peak. However, notice that the instrumentation used to create the diagrams is actually plotting *load versus deflection* rather than *true stress versus strain*. The apparent stress is computed by dividing the load by the original cross-sectional area of the test bar. After the peak of the curve is reached, there is a pronounced decrease in the bar's diameter, referred to as *necking down*. Thus,

the load acts over a smaller area, and the *actual stress* continues to increase until failure. It is very difficult to follow the reduction in diameter during the necking-down process, so it has become customary to use the peak of the curve as the tensile strength, although it is a more conservative value.

Yield Strength, s_y

That portion of the stress-strain diagram where there is a large increase in strain with little or no increase in stress is called the *yield strength* (s_y). This property indicates that the material has, in fact, yielded or elongated plastically, permanently, and to a large degree. If the point of yielding is quite noticeable, as it is in Figure 2-1, the property is called the *yield point* rather than the yield strength. This is typical of a plain carbon hot-rolled steel.

Figure 2-2 shows the stress-strain diagram that is typical of a nonferrous metal such as aluminum or titanium or of certain high-strength steels. Notice that there is no pronounced yield point, but the material has actually yielded at or near the stress level indicated as s_y . That point is determined by the *offset method*, in which a line is drawn parallel to the straight-line portion of the curve and is offset to the right by a set amount, usually 0.20% strain (0.002 in/in). The intersection of this line and the stress-strain curve defines the material's yield strength. In this book, the term *yield strength* will be used for s_y , regardless of whether the material exhibits a true yield point or whether the offset method is used.

Proportional Limit

That point on the stress-strain curve where it deviates from a straight line is called the *proportional limit*. That is, at or above that stress value, stress is no longer proportional to strain. Below the proportional limit, Hooke's law applies: Stress is proportional to strain. In mechanical design, materials are rarely used at stresses above the proportional limit.

Elastic Limit

At some point, called the *elastic limit*, a material experiences some amount of plastic strain and thus will not return to its original shape after release of the load. Below that level, the material behaves completely elastically. The proportional limit and the elastic limit lie quite close to the yield strength. Because they are difficult to determine, they are rarely reported.

Modulus of Elasticity in Tension, E

For the part of the stress-strain diagram that is straight, stress is proportional to strain, and the value of E , the *modulus of elasticity*, is the constant of proportionality. That is,

Modulus of Elasticity in Tension

$$E = \frac{\text{stress}}{\text{strain}} = \frac{\sigma}{\epsilon} \quad (2-1)$$

This is the slope of the straight-line portion of the diagram. The modulus of elasticity indicates the stiffness of the material, or its resistance to deformation.

Ductility and Percent Elongation

Ductility is the degree to which a material will deform before ultimate fracture. The opposite of ductility is *brittleness*. When ductile materials are used in machine members, impending failure is detected easily, and sudden failure is unlikely. Also, ductile materials normally resist the repeated loads on machine elements better than brittle materials.

The usual measure of ductility is the *percent elongation* of the material after fracture in a standard tensile test. Figure 2-3 shows a typical standard tensile specimen before and after the test. Before the test, gage marks are placed on the bar, usually 2.00 in apart. Then, after the bar is broken, the two parts are fitted back together, and the final length between the gage marks is measured. The percent elongation is the difference between the final length and the original length divided by the original length, converted to a percentage. That is,

Percent Elongation

$$\text{Percent elongation} = \frac{L_f - L_o}{L_o} \times 100\% \quad (2-2)$$

The percent elongation is assumed to be based on a gage length of 2.00 in unless some other gage length is specifically indicated. Tests of structural steels often use a gage length of 8.00 in.

Theoretically, a material is considered ductile if its percent elongation is greater than 5% (lower values indicate brittleness). For practical reasons, it is advisable to use a material with a value of 12% or higher for machine members subject to repeated loads or shock or impact.

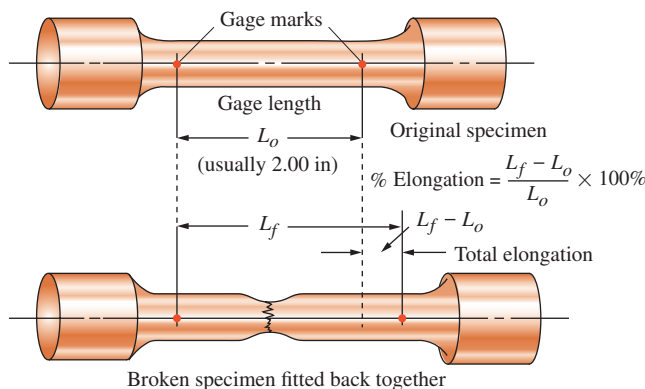


FIGURE 2-3 Measurement of percent elongation

Percent reduction in area is another indication of ductility. To find this value, compare the original cross-sectional area with the final area at the break for the tensile test specimen.

Shear Strength, s_{ys} and s_{us}

Both the yield strength and the ultimate strength in shear (s_{ys} and s_{us} , respectively) are important properties of materials. Unfortunately, these values are seldom reported. We will use the following estimates:

Estimates for s_{ys} and s_{us}

$$s_{ys} = s_y/2 = 0.50 s_y = \text{yield strength in shear} \quad (2-3)$$

$$s_{us} = 0.75 s_u = \text{ultimate strength in shear} \quad (2-4)$$

Poisson's Ratio, ν

When a material is subjected to a tensile strain, there is a simultaneous shortening of the cross-sectional dimensions perpendicular to the direction of the tensile strain. The ratio of the shortening strain to the tensile strain is called *Poisson's ratio*, usually denoted by ν , the Greek letter nu. (The Greek letter mu, μ , is sometimes used for this ratio.) Poisson's ratio is illustrated in Figure 2-4. Typical ranges of values for Poisson's ratio are shown in Table 2-1.

Modulus of Elasticity in Shear, G

The *modulus of elasticity in shear* (G) is the ratio of shearing stress to shearing strain. This property indicates a material's stiffness under shear loading—that is, the resistance to shear deformation. There is a simple relationship between E , G , and Poisson's ratio:

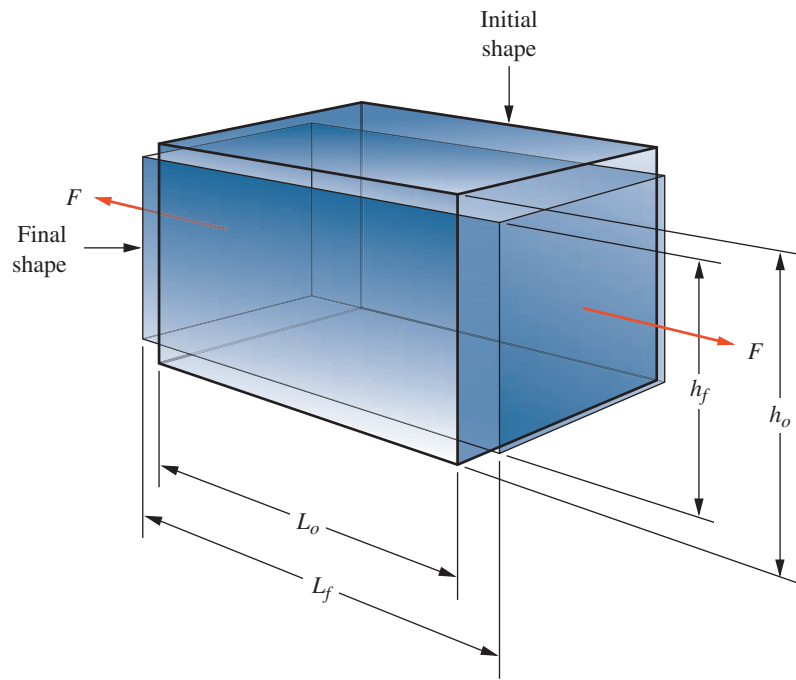
Modulus of Elasticity in Shear

$$G = \frac{E}{2(1 + \nu)} \quad (2-5)$$

This equation is valid within the elastic range of the material.

Non-Destructive Measurement of Elastic Modulus and Poisson's Ratio

While most determinations for the elastic modulus in tension and in shear and the related property of Poisson's ratio are done using mechanical destructive testing as previously described, a nondestructive method exists that can be used for most metals, ceramics, and glasses. Internet sites 41 describes a method that uses a measurement of the velocity of sound in the material using an ultrasonic pulse-echo technique along with a thickness measuring device. A sample of material at least 12 mm (0.50 in) thick with smooth parallel surfaces is used. After measuring the sound velocity,

FIGURE 2-4 Illustration of Poisson's ratio for an element in tension

$$\text{Axial strain} = \frac{L_f - L_o}{L_o} = \epsilon_a$$

$$\text{Lateral strain} = \frac{h_f - h_o}{h_o} = \epsilon_L$$

$$\text{Poisson's ratio} = \frac{-\epsilon_L}{\epsilon_a} = \nu$$

TABLE 2-1 Approximate Values of Poisson's Ratio, ν

Concrete	0.10–0.25	Aluminum (most alloys)	0.33
Glass	0.24	Brass	0.33
Ductile iron	0.27	Copper	0.33
Gray cast iron	0.21	Zinc	0.33
Plastics	0.20–0.40	Phosphor bronze	0.35
Carbon and alloy steel	0.29	Magnesium	0.35
Stainless steel (18-8)	0.30	Lead	0.43
Titanium	0.30	Rubber, elastomers	~0.50

Notes: Values are approximate and vary somewhat with specific composition. Rubber and elastomers approach a limiting value of 0.50.

simple calculations enable determination of Poisson's ratio, Young's modulus (elastic modulus in tension), and shear modulus.

Flexural Strength and Flexural Modulus

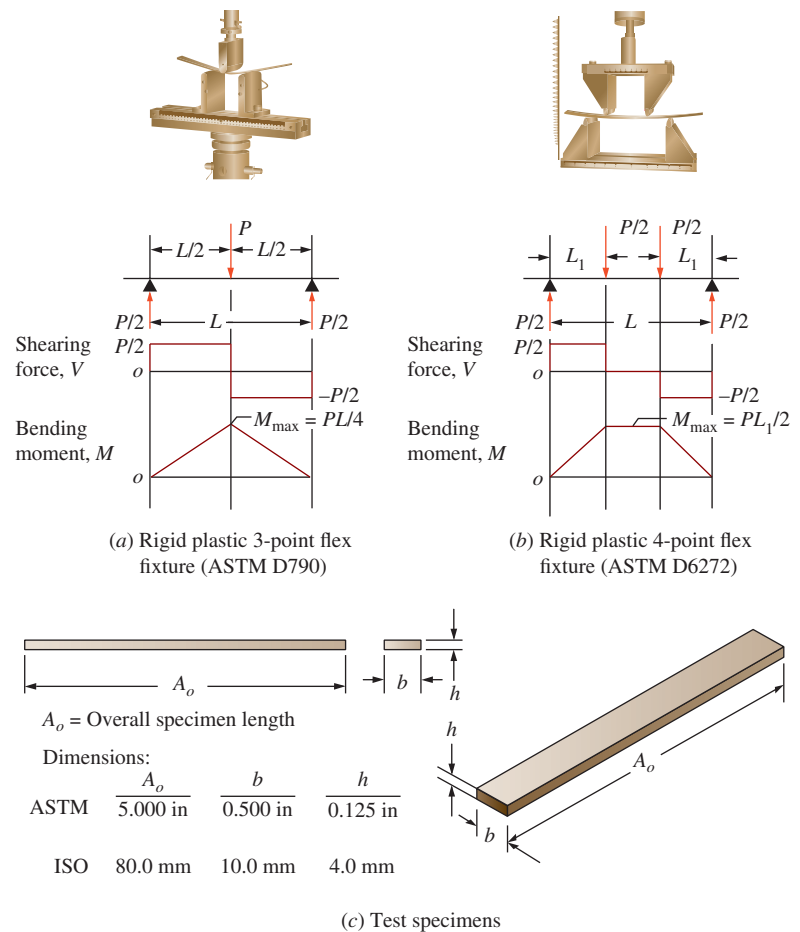
For some materials, particularly plastics, additional properties in bending are obtained, called *flexural strength* and *flexural modulus*. Typical testing fixtures are shown in Figure 2-5 and testing should be done according to the recognized standards from either ASTM or ISO. ASTM D790¹ and ISO 178² use the 3-point bending principle [Part (a) of Figure 2-5], wherein a flat specimen is placed

on two cylindrical or knife-edge supports with a known span, L , between them. The bending force is applied at mid-span ($L/2$). The specimen for plastic materials is typically a flat strip with the dimensions shown. Its section modulus, S , is needed for bending stress calculations. From Appendix 1, $S = bh^2/6$. The maximum

¹ASTM International. *Standard Test Method for Flexural Properties of Unreinforced and Reinforced Plastics and Electrical Insulating Materials*, Standard D790. West Conshohocken, PA: ASTM International, DOI: 10.1520/D790-10, www.astm.org, 2010.

²International Standards Organization (ISO). *Plastics—Determination of Flexural Properties*. ISO 178:2001.

FIGURE 2-5 Test fixtures for flexural testing



bending moment occurs at the point of load application and has the value, $M_{\max} = PL/4$. Then the flexural (bending) stress is:

$$\sigma = \frac{M_{\max}}{S} = \frac{PL/4}{bh^2/6} = \frac{1.5PL}{bh^2} \quad (2-6)$$

The stress at break or after a certain specified percentage of deflection is reported as the flexural strength. One disadvantage of this method is that shearing stress exists along with bending stress and that may affect the results for some specimens.

Flexural modulus is measured as the slope of the load-deflection curve taken during the test at the straight-line portion, if any. Some plastic materials exhibit a nonlinear slope for the curve and other techniques should be used. One uses a *secant* line drawn between two agreed-upon points on the curve.

The 4-point bending principle, as used in ASTM D6272³ and ISO 14125⁴, overcomes the shearing stress issue described above because the highest stressed part of the beam experiences no shearing force. See Part (b)

of Figure 2-5. The distances, L_1 , from the supports to each load are the same. Then, $M_{\max} = PL_1/2$ and the flexural stress is:

$$\sigma = \frac{M_{\max}}{S} = \frac{PL_1/2}{bh^2/6} = \frac{3PL_1}{bh^2} \quad (2-7)$$

Hardness

The resistance of a material to indentation by a penetrator is an indication of its *hardness*. Hardness in a steel indicates wear resistance as well as strength. Wear resistance will be discussed in later chapters, particularly with regard to gear teeth. Several types of devices, procedures, and penetrators measure hardness; the Brinell hardness tester and the Rockwell hardness tester are most frequently used for machine elements. For steels, the Brinell hardness tester employs a hardened steel ball 10 mm in diameter as the penetrator under a load of 3000-kg force. The load causes a permanent indentation in the test material, and the diameter of the indentation is related to the Brinell hardness number, which is abbreviated BHN or HB. The actual quantity being measured is the load divided by the contact area of the indentation. For steels, the value of HB ranges from approximately 100 for an annealed, low-carbon steel to more than 700 for high-strength, high-alloy steels in the as-quenched condition. In the high ranges, above HB 500,

³ASTM International. *Standard Test Method for Flexural Properties of Unreinforced and Reinforced Plastics and Electrical Insulating Materials by Four-Point Bending*, Standard D6272. West Conshohocken, PA: ASTM International, DOI: 10.1520/D6272-10, www.astm.org, 2010.

⁴International Standards Organization (ISO). *Fibre-Reinforced Plastic Composites—Determination of Flexural Properties*. ISO 14125:1998/Cor.1:2001(E).

the penetrator is sometimes made of tungsten carbide rather than steel. For softer metals, a 500-kg load is used.

The Rockwell hardness tester uses a hardened steel ball with a 1/16-in diameter under a load of 100-kg force for softer metals, and the resulting hardness is listed as Rockwell B, R_B , or HRB. For harder metals, such as heat-treated alloy steels, the Rockwell C scale is used. A load of 150-kg force is placed on a diamond penetrator (a *brale* penetrator) made in a sphero-conical shape. Rockwell C hardness is sometimes referred to as R_C or HRC. Many other Rockwell scales are used.

The Brinell and Rockwell methods are based on different parameters and lead to quite different numbers. However, since they both measure hardness, there is a correlation between them, as noted in Appendix 17 and Figure 2-6. It is also important to note that, especially for highly hardenable alloy steels, there is a nearly linear relationship between the Brinell hardness number and the tensile strength of the steel, according to the equation

➤ **Approximate Relationship between Hardness and Strength for Steel**

$$0.50(\text{HB}) = \text{approximate tensile strength (ksi)} \quad (2-8)$$

This relationship is shown in Figure 2-6.

To compare the hardness scales with the tensile strength, consider Table 2-2. Note that there is some overlap between the HRB and HRC scales. Normally, HRB is used for the softer metals and ranges from approximately 60 to 100, whereas HRC is used for harder metals and ranges from 20 to 65. Using HRB numbers above 100 or HRC numbers below 20 is not recommended. Those shown in Table 2-2 are for comparison purposes only.

The Vickers hardness test, preferred in some European countries and elsewhere, is similar to the Brinell test except for the nature of the penetrator (square-based diamond pyramid) and the applied load, typically 50 kg. Appendix 17 includes the Vickers hardness for comparison with Brinell and Rockwell hardness values. See also Reference 38.

Whereas Rockwell B and C scales are used for substantial components made from steels, other Rockwell hardness scales can be used for different materials as listed in Table 2-3.

For plastics, rubbers, and elastomers, it is typical to use the Shore or IRHD (International rubber hardness degree) methods. Table 2-4 lists some of the more

FIGURE 2-6 Hardness conversions

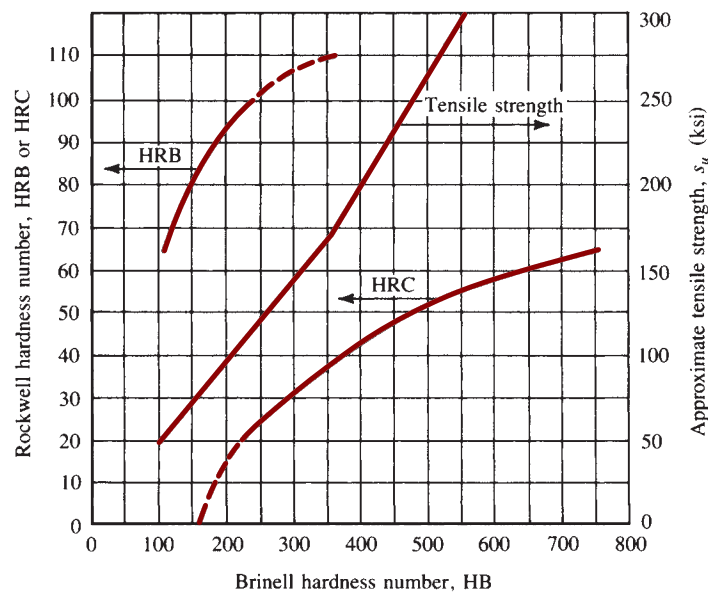


TABLE 2-2 Comparison of Hardness Scales with Tensile Strength

Material and condition	Hardness			Tensile strength	
	HB	HRB	HRC	ksi	MPa
1020 annealed	121	70		60	414
1040 hot-rolled	144	79		72	496
4140 annealed	197	93	13	95	655
4140 OQT 1000	341	109	37	168	1160
4140 OQT 700	461		49	231	1590

TABLE 2–3 Rockwell Hardness Scales and Their Uses

Scale	Symbol	Typical uses
A	HRA	Thin steel, hardened steel with shallow case
B	HRB	Softer steels such as low carbon, annealed or hot-rolled; softer aluminum, copper, cast irons
C	HRC	Harder steels such as heat-treated alloy steels, tool steels, titanium
D	HRD	Medium-depth case-hardened steels, harder cast irons
E	HRE	Harder bearing metals, aluminum, magnesium, cast irons
F	HRF	Soft thin sheet metals, annealed copper and zinc alloys
G	HRG	Beryllium copper, phosphor bronze, softer cast irons
H	HRH	Aluminum, zinc, lead
K	HRK	Softer bearing metals, plastics, rubbers, and other soft materials. Scales L, M, P, R, S, V, and α also available
15 N	HR15N	Similar to A scale but for thinner materials or thin case hardening
30 N	HR30N	Similar to C scale but for thinner materials or thin case hardening
45 N	HR45N	Similar to D scale but for thinner materials or thin case hardening
15 T	HR15T	Similar to B scale but for thinner materials
30 T	HR30T	Similar to F scale but for thinner materials
45 T	HR45T	Similar to G scale but for thinner materials

Notes: Measurement devices use different indenter sizes and shapes and applied forces. For details, consult standards ASTM E18, ISO 6508.

TABLE 2–4 Hardness Measurement for Plastics, Rubbers, and Elastomers

Shore method		
Scale of durometer	Symbol	Typical uses
A	Shore A	Soft vulcanized natural rubbers, elastomers (e.g., neoprene), leather, wax, felt
B	Shore B	Moderately hard rubber as used for printer rolls and platens
C	Shore C	Medium hard rubbers and plastics
D	Shore D	Hard rubbers and plastics such as vinyl sheets, Plexiglas, laminate countertops
DO	Shore DO	Very dense textile windings
O	Shore O	Soft rubber such as artgum; textile windings
OO	Shore OO	Low density textile windings; sponge rubber
OOO	Shore OOO	Soft plastic foams
T	Shore T	Medium density textiles on spools
M	Shore M	Rubber O-rings and thin sheet rubber
IRHD method (International rubber hardness degree)		
Name	Typical uses	
IRHD Micro	Small: O-rings, small components, thin materials	
IRHD Macro L	Soft: Readings up to 35 IRHD L	
IRHD Macro N	Normal: Readings from 30 IRHD N to 100 IRHD N	

Notes: Measurement devices use different indenter sizes and shapes and applied forces. For details, consult standards ASTM D2240, ISO 868, or DIN 53505.

popular scales. Also used for some plastics are the Rockwell R, L, M, E, K, and α scales. These vary by the size and geometry of the indenter and the applied force.

Other hardness measurement methods include Knoop, Universal, and rebound hardness.

Wear in Mechanical Devices

Wear is an issue whenever two components operate with relative motion between them or when liquids or solids impinge on a surface at high velocity. The action of one surface on another tends to develop material loss, scoring, or roughening of the mating surfaces. Wear is often cumulative and can eventually render the components incapable of delivering their expected performance. See References 4, 7, 24, and 34. Failure modes include developing:

- Excessive clearances between the mating objects
- Noise
- Unacceptable appearance of either mating surface
- Increased friction at the mating surfaces
- Progressive rise in temperature at the mating surfaces
- Seizure resulting in complete stoppage of the device

Designers must take steps to minimize wear by proper material selection, surface preparation during manufacture, lubrication of the mating surfaces, periodic inspection, and remedial steps to overcome the effects of wear in early stages.

While not really a material property itself, it is pertinent to mention wear in this chapter because the characteristics of the materials at the mating surfaces determine the potential for wear to occur and to what degree. Conditions that the designer must control include:

- Chemical compatibility of the materials of the two contacting surfaces, of any lubricant present, and the environment in which the equipment is to operate to inhibit corrosion that would build up on mating surfaces, cause roughening, increase friction, and initiate material removal.
- The hardness of the materials at the contacting surfaces; typically harder materials have greater ability to resist wear, although exceptions occur.
- The smoothness of the surface finish at areas of contact to minimize the tendency for microscopic hills and valleys to come into contact.
- The physical size of the mating area over which applied forces act to control the effective pressure between mating bodies to a manageable level.
- The inherent coefficient of friction between the mating surfaces at operating conditions.
- The type of lubricant used, if any, and its ability to effectively separate the sliding bodies while under load.
- Control of the temperature of the mating components and of any lubricant used because elevated

temperatures can reduce the effectiveness of the lubricant and lower the strength of the materials in contact.

- The cleanliness of the lubricant and of the surfaces themselves because even small solid particles between the mating surfaces can initiate scoring and roughening of the surfaces.

Several kinds of wear can exist.

- **Erosive wear**—The displacement of particles from a surface due to impact of solids or liquids on a surface as might occur on equipment subjected to wind and rain or the insides of pipes, elbows, and other fittings carrying liquids or gases containing solids, called slurries.
- **Abrasive wear**—The mechanical tearing of particles from one material by the action of the mating material resulting in the loss of mass from one or both materials.
- **Adhesive wear**—The tendency of one material to adhere to the mating material and subsequently remove particles by breaking the adhesion and making the surfaces rougher.
- **Fretting wear**—Cyclical relative motion of two tightly joined parts under high surface pressure as might occur with connectors, fasteners, clamps, and similar situations where even vibration can initiate fretting wear.
- **Surface fatigue**—Progressive damage caused by creating high contact stresses between mating components that eventually lead to fatigue failure of one of the mating components. Examples include the action of rolling balls or rollers on the inner or outer races of bearings, treads of wheels rolling on rails or flat surfaces, and supports for beams where high stresses and relative motion can occur.

It is virtually impossible to predict the amount of wear that may occur in a given situation and testing of the proposed design is recommended. At times, designers can benefit from test data from samples of materials under controlled loads and mating conditions. At least a measure of relative merit or ranking of certain combinations of materials can be determined. Standardized testing can be done using either of two methods described by ASTM International (See Internet site 5):

1. ASTM Standard G65-15 Standard Test Method for Measuring Abrasion Using the Dry Sand/Rubber Wheel Apparatus, DOI: 10.1520/G0065-15.
 - a. Conditions are standardized for abrasive particle size, shape, and hardness; the magnitude of the stress imposed by the particle; and the frequency of contact of the abrasive particles to create scratching abrasion. Volume loss from the test materials is measured.
2. ASTM Standard G132-2013 Standard Test Method for Pin Abrasion Testing, DOI: 10.1520/G0132-13.
 - a. Relative motion is established between flat surfaces of two materials while maintaining a measurable force between them. Mass loss from the test materials is measured.

Machinability

Machinability is related to the ease with which a material can be machined to a good surface finish with reasonable tool life. Production rates are directly affected by machinability. It is difficult to define measurable properties related to machinability, so machinability is usually reported in comparative terms, relating the performance of a given material with some standard.

Toughness, Impact Energy

Toughness is the ability of a material to absorb applied energy without failure. Parts subjected to suddenly applied loads, shock, or impact need a high level of toughness. Several methods are used to measure the amount of energy required to break a particular specimen made from a material of interest. The energy absorption value from such tests is often called *impact energy* or *impact resistance*. However, it is important to note that the actual value is highly dependent on the nature of the test sample, particularly its geometry. It is not possible to use the test results in a quantitative way when making design calculations. Rather, the impact energy for several candidate materials for a particular application can be compared with each other as a qualitative indication of their toughness. The final design should be tested under real service conditions to verify its ability to survive safely during expected use.

For metals and plastics, two methods of determining impact energy, *Izod* and *Charpy*, are popular, with data often reported in the literature from vendors of the material. Figure 2–7 shows sketches of the dimensions of standard specimens and the manner of loading. In

each method, a pendulum with a heavy mass carrying a specially designed striker is allowed to fall from a known height. The striker contacts the specimen with a high velocity at the bottom of the pendulum's arc; therefore, the pendulum possesses a known amount of kinetic energy. The specimen is typically broken during the test, taking some of the energy from the pendulum but allowing it to pass through the test area. The testing machine is configured to measure the final height to which the pendulum swings and to indicate the amount of energy removed. That value is reported in energy units of J (Joules or $\text{N} \cdot \text{m}$) or $\text{ft} \cdot \text{lb}$. Some highly ductile metals and many plastics do not break during the test, and the result is then reported as *No Break*.

The standard *Izod* test employs a square specimen with a V-shaped notch carefully machined 2.0 mm (0.079 in) deep according to specifications in ASTM standard D 256.⁵ The specimen is clamped in a special vise with the notch aligned with the top edge of the vise. The striker contacts the specimen at a height of 22 mm above the notch, loading it as a cantilever in bending. When used for plastics, the width dimension can be different from that shown in Figure 2–7. This obviously changes the total amount of energy that the specimen will absorb during fracture. Therefore, the data for impact energy are divided by the actual width of the specimen, and the results are reported in units of $\text{N} \cdot \text{m}/\text{m}$ or $\text{ft} \cdot \text{lb}/\text{in}$. Also, some vendors and customers may agree to test the material with the

⁵ASTM International. *Standard Test Methods for Determining the Izod Pendulum Impact Resistance of Plastics, Standard D256-10e1*. West Conshohocken, PA: ASTM International, DOI: 10.1520/D0256-10, 2010.

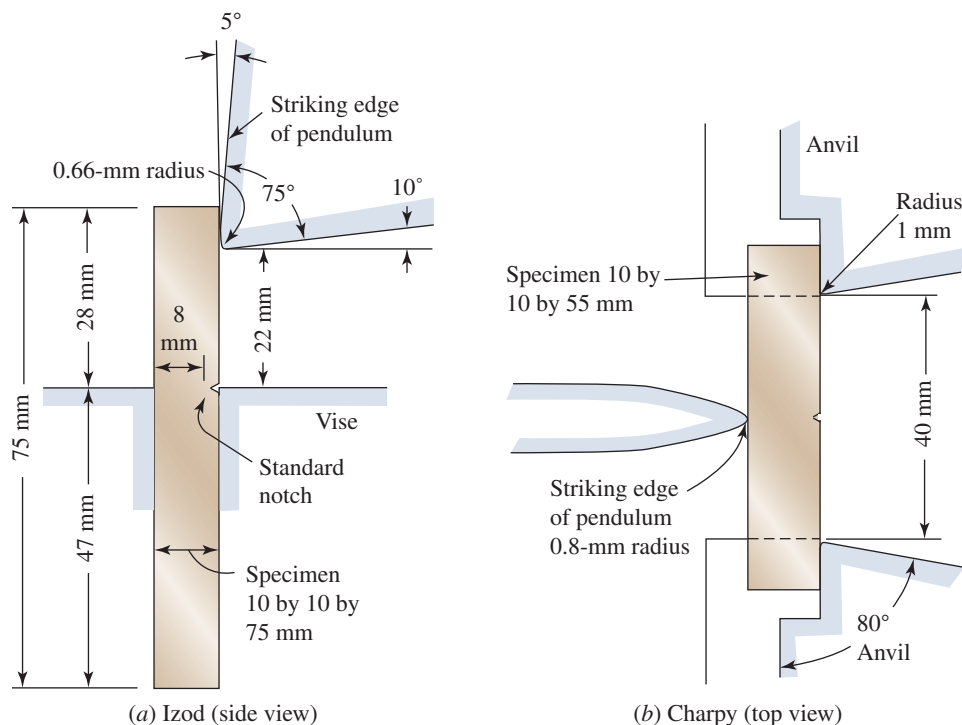


FIGURE 2–7 Impact testing using Charpy and Izod methods

notch facing away from the striker rather than toward it as shown in Figure 2–7. This gives a measure of the material's impact energy with less influence from the notch.

The *Charpy* test also uses a square specimen with a 2.0 mm (0.079 in) deep notch, but it is centered along the length. The specimen is placed against a rigid anvil without being clamped. See ASTM standard A 370⁶ for the specific geometry and testing procedure. The notch faces away from the place where the striker contacts the specimen. The loading can be described as the bending of a simply supported beam. The Charpy test is most often used for testing metals.

Another impact testing method used for some plastics, composites, and completed products is the *drop-weight* tester. Here a known mass is elevated vertically above the test specimen to a specified height. Thus, it has a known amount of potential energy. Allowing the mass to fall freely imparts a predictable amount of kinetic energy to the specimen clamped to a rigid base. The initial energy, the manner of support, the specimen geometry, and the shape of the striker (called a *tup*) are critical to the results found. One standard method, described in ASTM D 3763,⁷ employs a spherical tup with a diameter of 12.7 mm (0.50 in). The tup usually pierces the specimen. The apparatus is typically equipped with sensors that measure and plot the load versus deflection characteristics dynamically, giving the designer much information about how the material behaves during an impact event. Summary data reported typically include maximum load, deflection of the specimen at the point of maximum load, and the energy dissipated up to the maximum load point. The energy is calculated by determining the area under the load-deflection diagram. The appearance of the test

specimen is also described, indicating whether fracture occurred and whether it was a ductile or brittle fracture.

Fatigue Strength or Endurance Strength

Parts subjected to repeated applications of loads or to stress conditions that vary with time over several thousands or millions of cycles fail because of the phenomenon of *fatigue*. Materials are tested under controlled cyclic loading to determine their ability to resist such repeated loads. The resulting data are reported as the *fatigue strength*, also called the *endurance strength* of the material. (See Chapter 5.)

Creep

When materials are subjected to high loads continuously, they may experience progressive elongation over time. This phenomenon, called *creep*, should be considered for metals operating at high temperatures. You should check for creep when the operating temperature of a loaded metal member exceeds approximately 0.3 (T_m) where T_m is the melting temperature expressed as an absolute temperature. (See Reference 25.) Creep can be important for critical members in internal combustion engines, furnaces, steam turbines, gas turbines, nuclear reactors, or rocket engines. The stress can be tension, compression, flexure, or shear. (See Reference 12.)

Figure 2–8 shows the typical behavior of metals that creep. The vertical axis is the creep strain, in units such as in/in or mm/mm, over that which occurs initially as the load is applied. The horizontal axis is time, typically measured in hours because creep develops slowly over a long term. During the primary portion of the creep strain versus time curve, the rate of increase in strain initially rises with a rather steep slope that then decreases. The slope is constant (straight line) during the secondary portion of the curve. Then the slope increases in the tertiary portion that precedes the ultimate fracture of the material.

Creep is measured by subjecting a specimen to a known steady load, possibly through application of a

⁶ASTM International. *Standard Test Methods and Definitions for Mechanical Testing of Steel Products, Standard A370-15*. West Conshohocken, PA: ASTM International, DOI: 10.1520/A0370-2015.

⁷ASTM International. *Standard Test Methods for High Speed Puncture of Plastics Using Load and Displacement Sensors, Standard D3763-15*. West Conshohocken, PA: ASTM International, DOI: 10.1520/D3763-2015.

FIGURE 2–8 Typical creep behavior

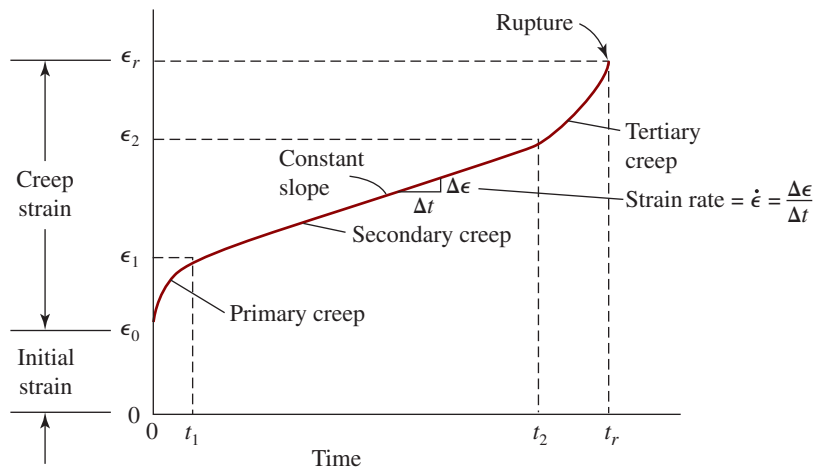
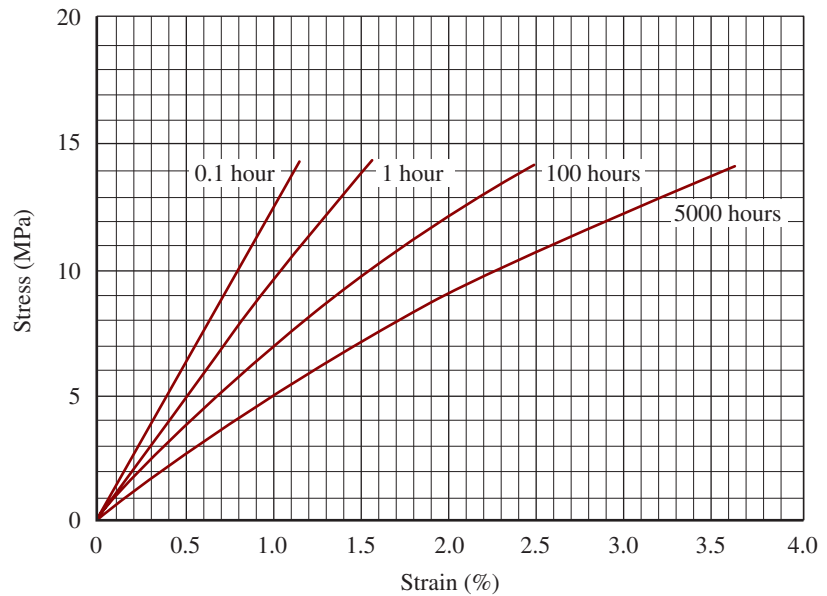


FIGURE 2-9 Example of stress versus strain as a function of time for nylon 66 plastic at 23°C (73°F)



dead weight, while the specimen is heated and maintained at a uniform temperature. Data for strain versus time are taken at least into the secondary creep stage and possibly all the way to fracture to determine the creep rupture strain. Testing over a range of temperatures gives a family of curves that are useful for design.

Creep can occur for many plastics even at or near room temperature. Figure 2-9 shows one way that creep data are displayed for plastic materials. (See References 14.) It is a graph of applied stress versus strain in the member with data shown for a specific temperature of the specimen. The curves show the amount of strain that would be developed within the specified times at increasing stress levels. For example, if this material were subjected to a constant stress of

5.0 MPa for 5000 hours, the total strain would be 1.0%. That is, the specimen would elongate by an amount 0.01 times the original length. If the stress were 10.0 MPa for 5000 hours, the total strain would be approximately 2.25%. The designer must take this creep strain into account to ensure that the product performs satisfactorily over time.

Relaxation

A phenomenon related to creep occurs when a member under stress is captured under load, giving it a certain fixed length and a fixed strain. Over time, the stress in the member would decrease, exhibiting a behavior called *relaxation*. This is important in such applications

Example Problem 2-1

A solid circular bar has a diameter of 5.0 mm and a length of 250 mm. It is made from nylon 66 plastic (30% Glass, 50% R.H.) and subjected to a steady tensile load of 240 N. Compute the elongation of the bar immediately after the load is applied and after 5000 hours (approximately seven months). See Appendix 13 and See Figure 2-9 for properties of the nylon.

Solution

The stress and deflection immediately after loading will first be computed using fundamental equations of strength of materials:

$$\sigma = F/A \text{ and } \delta = FL/EA$$

See Chapter 3 for a review of strength of materials.

Then creep data from Figure 2-9 will be applied to determine the elongation after 5000 hours.

Results

Stress:

The cross-sectional area of the bar is

$$A = \pi D^2/4 = \pi(5.0 \text{ mm})^2/4 = 19.63 \text{ mm}^2$$

$$\sigma = \frac{F}{A} = \frac{240 \text{ N}}{19.63 \text{ mm}^2} = 12.2 \text{ N/mm}^2 = 12.2 \text{ MPa}$$

Appendix 13 lists the tensile strength for nylon 66 to be 102 MPa. Therefore, the rod is safe from fracture.

Elongation:

The tensile modulus of elasticity for nylon 66 is found from Appendix 13 to be $E = 5500$ MPa. Then the initial elongation is,

$$\delta = \frac{FL}{EA} = \frac{(240 \text{ N})(250 \text{ mm})}{(5500 \text{ N/mm}^2)(19.63 \text{ mm}^2)} = 0.556 \text{ mm}$$

Creep:

Referring to Figure 2–9 we find that when a tensile stress of 12.2 MPa is applied to the nylon 66 plastic for 5000 hours, a total strain of approximately 2.95% occurs. This can be expressed as

$$\epsilon = 2.95\% = 0.0295 \text{ mm/mm} = \delta/L$$

Then,

$$\delta = \epsilon L = (0.0295 \text{ mm/mm})(250 \text{ mm}) = 7.375 \text{ mm}$$

Comment This is approximately seven times as much deformation as originally experienced when the load was applied. So designing with the reported value of modulus of elasticity is not appropriate when stresses are applied continuously for a long time. We can now compute an apparent modulus of elasticity, E_{app} , for this material at the 5000 hours service life.

$$E_{\text{app}} = \sigma/\epsilon = (12.2 \text{ MPa})/(0.0295 \text{ mm/mm}) = 414 \text{ MPa}$$

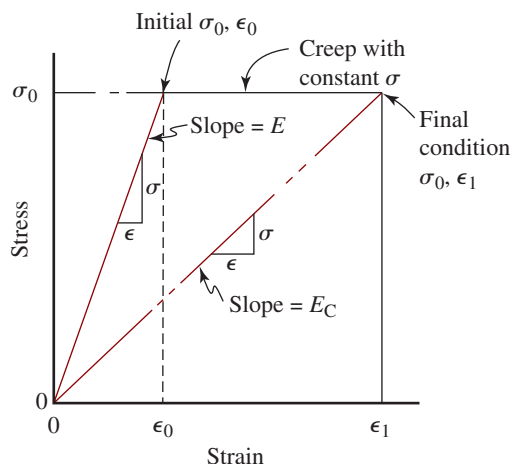
as clamped joints, press-fit parts, and springs installed with a fixed deflection. Figure 2–10 shows the comparison between creep and relaxation. For stresses below approximately 1/3 of the ultimate tensile strength of the material at any temperature, the apparent modulus in either creep or relaxation at any time of loading may be considered similar for engineering purposes. Furthermore, values for apparent modulus are the same for tension, compression, or flexure. (See Reference 14.) Analysis of relaxation is complicated by the fact that as the stress decreases, the rate of creep also decreases. Additional material data beyond that typically reported would be required to accurately predict the amount of relaxation at any given time. Testing under realistic conditions is recommended.

Physical Properties

Here we will discuss density, coefficient of thermal expansion, thermal conductivity, and electrical resistivity.

Density. *Density* is defined as the mass per unit volume of a material. Its usual units are kg/m^3 in the SI and lb/in^3 in the U.S. Customary Unit System, where the pound unit is taken to be pounds-mass. The Greek letter rho (ρ) is the symbol for density.

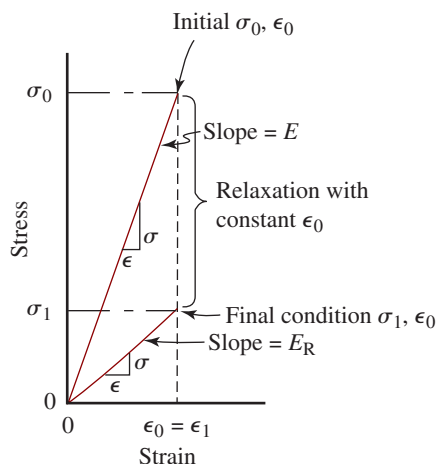
In some applications, the term *specific weight* or *weight density* is used to indicate the weight per unit volume of a material. Typical units are N/m^3 in the SI and lb/in^3 in the U.S. Customary Unit System, where the pound is taken to be pounds-force. The Greek letter gamma (γ) is the symbol for specific weight.



$$E = \frac{\sigma_0}{\epsilon_0} = \text{Tensile modulus}$$

$$E_C = \frac{\sigma_0}{\epsilon_1} = \text{Creep modulus}$$

(a) Creep behavior



$$E = \frac{\sigma_0}{\epsilon_0} = \text{Tensile modulus}$$

$$E_R = \frac{\sigma_1}{\epsilon_0} = \text{Relaxation modulus}$$

(b) Relaxation behavior

FIGURE 2–10 Comparison of creep and relaxation

Coefficient of Thermal Expansion. The *coefficient of thermal expansion* is a measure of the change in length of a material subjected to a change in temperature. It is defined by the relation

⇒ **Coefficient of Thermal Expansion**

$$\alpha = \frac{\text{change in length}}{L_o(\Delta T)} = \frac{\text{strain}}{(\Delta T)} = \frac{\epsilon}{(\Delta T)} \quad (2-9)$$

where L_o = original length

ΔT = change in temperature

Virtually all metals and plastics expand with increasing temperature, but different materials expand at different rates. For machines and structures containing parts of more than one material, the different rates can have a significant effect on the performance of the assembly and on the stresses produced.

Thermal Conductivity. *Thermal conductivity* is the property of a material that indicates its ability to transfer heat. Where machine elements operate in hot environments or where significant internal heat is generated, the ability of the elements or of the machine's housing to transfer heat away can affect machine performance. For example, wormgear speed reducers typically generate frictional heat due to the rubbing contact between the worm and the wormgear teeth. If not adequately transferred, heat causes the lubricant to lose its effectiveness, allowing rapid gear-tooth wear.

Electrical Resistivity. For machine elements that conduct electricity while carrying loads, the electrical resistivity of the material is as important as its strength. *Electrical resistivity* is a measure of the resistance offered by a given thickness of a material; it is measured in ohm-centimeters ($\Omega \cdot \text{cm}$). *Electrical conductivity*, a measure of the capacity of a material to conduct electric current, is sometimes used instead of resistivity. It is often reported as a percentage of the conductivity of a reference material, usually the International Annealed Copper Standard.

2-3 CLASSIFICATION OF METALS AND ALLOYS

Various industry associations take responsibility for setting standards for the classification of metals and alloys. Each has its own numbering system, convenient to the particular metal covered by the standard. But this leads to confusion at times when there is overlap between two or more standards and when widely different schemes are used to denote the metals. Order has been brought to the classification of metals by the use of the Unified Numbering Systems (UNS) as defined in the Standard E 527-12, *Standard Practice for Numbering Metals and Alloys (UNS)*, by the American Society for Testing and Materials, or ASTM. (See Reference 23.) Besides listing materials under the control of ASTM itself, the UNS coordinates designations of the following:

The Aluminum Association (AA)

The American Iron and Steel Institute (AISI)

The Copper Development Association (CDA)

The Society of Automotive Engineers (SAE)

The primary series of numbers within UNS are listed in Table 2-5, along with the organization having responsibility for assigning numbers within each series.

Recent Developments in Steel Designations

The most-used type of material for machine elements in this book is steel. Section 2-5 presents significant information on the various types of steel and their conditions as it relates to design decisions. For many years, two organizations, AISI and SAE, have played major roles in cataloging steels and in specifying the designation systems used in the United States. This work was coordinated with steel producers, professional societies, and design engineers in several steel-using sectors of industry. To a large extent, the designation systems of AISI and SAE were identical. The automotive industry tended to use the SAE system while a broad array of other industries used the AISI system. In recent years, the AISI has discontinued the practice of assigning steel designation numbers, and the SAE now plays the lead role.

In parallel with these developments, standards-writing committees of SAE, in cooperation with ASTM, began a process of broadening the designation systems for wrought and rolled steels beyond the series listed in Table 2-5. New systems for series G, H, and K have been developed and are described in SAE Standard J402 (R) *New Steel Designation System for Wrought or Rolled Steel*. These new steel designations are the same in both the SAE and UNS systems as described in SAE Standard J402 and the joint SAE J1086/ASTM E527 publication. Many of the alloys listed within series G, H, and K retain the familiar four-digit AISI and SAE designations as a *part* of the UNS number. Details of the new steel designations will be discussed in Section 2-5.

Aerospace Materials System (AMS). The development of the AMS system for designating materials resulted from the needs for special materials to meet the unique conditions encountered in aerospace applications. Aerospace Materials Specifications (AMS), published by SAE, are complete specifications that are generally adequate for procurement purposes. Most of the AMS designations pertain to materials intended for aerospace applications. The specifications may include mechanical property requirements significantly more severe than those for grades of steel having similar compositions but intended for other applications. Processing requirements, such as for consumable electrode remelting, are common in AMS steels.

AMS standards can be accessed through Internet site 4 for SAE International by selecting the *Standards* tab, then *Standards by Industry*, then *Aerospace*. At that point, if you are looking for a standard for a particular steel alloy, use the keyword search and enter the SAE alloy designation. For example, searching on "SAE 4340" will bring up several standards for the application

TABLE 2-5 Unified Numbering System (UNS)

Number series	Types of metals and alloys	Responsible organization
Nonferrous metals and alloys		
A00001–A99999	Aluminum and aluminum alloys	AA
C00001–C99999	Copper and copper alloys	CDA
E00001–E99999	Rare earth metals and alloys	ASTM
L00001–L99999	Low-melting metals and alloys	ASTM
M00001–M99999	Miscellaneous nonferrous metals and alloys	ASTM
N00001–N99999	Nickel and nickel alloys	SAE
P00001–P99999	Precious metals and alloys	ASTM
R00001–R99999	Reactive and refractory metals and alloys	SAE
Z00001–Z99999	Zinc and zinc alloys	ASTM
Ferrous metals and alloys		
D00001–D99999	Steels; mechanical properties specified	SAE
F00001–F99999	Cast irons and cast steels	ASTM
¹ G00001–G99999	Carbon and alloy steels	SAE
¹ H00001–H99999	H-steels; specified hardenability	SAE
J00001–J99999	Cast steels (except tool steels)	ASTM
¹ K00001–K99999	Miscellaneous steels and ferrous alloys	SAE
S00001–S99999	Heat- and corrosion-resistant (stainless) steels	ASTM
T00001–T99999	Tool steels	SAE

Note: ¹ The series G, H, and K systems have been changed as described in Section 2-5. For a time, both the form listed in this table and the new system may be seen in practice.

of this alloy with SAE 4340 in red. Standards deal with this material in bars, forgings, tubing, sheet, strip, plate, welding wire, castings, and several more.

Note that the lists of SAE and AMS standards on the SAE website include many other types of materials: *Adhesives, Sealants, Fabrics, Ceramics, Composites, Foams, Polymers*, and many more.

Material Property Data Used in This Book

The following sections of this chapter provide much additional detail about the types of metals, plastics, and composites for which data are given in the appendix and which are used in example problems, problems for your solution, and design projects. Here, we give an overview of those data and you are advised to peruse the appendix tables and those included in this section to become acquainted with the type of data available. This extensive set of data provides you with many options when selecting a material for a given design and gives you experience comparing their properties.

Appendices

- A-3 Properties of carbon and alloy steels
- A-4 Properties of heat-treated steels
- A-5 Properties of carburized steels
- A-6 Properties of stainless steels

- A-7 Properties of structural steels
- A-8 Properties of cast iron
- A-9 Properties of aluminum
- A-10 Properties of die cast zinc and magnesium alloys
- A-11 Properties of nickel-based alloys and titanium alloys
- A-12 Properties of bronze, brass, and other copper alloys
- A-13 Properties of selected plastics

Typical information and property data given for the materials are:

- Material name, type, and condition
 - For metals, conditions are typically hot-rolled, cold-drawn, annealed, normalized, or heat-treated. It is essential to specify the condition in order to fully predict the properties
- Tensile strength; sometimes called ultimate strength
 - For plastics, flexural strength and flexural modulus are also given
- Yield strength
- Percent elongation, used as a measure of the ductility of metals
- Impact strength for some metals and plastics
- Physical property data: modulus of elasticity, density; sometimes given in footnotes

Other pertinent data are given for particular types of materials.

Metals Used for Commercially Available Shapes

Many designs for components of machines are fabricated from standard forms produced by metal suppliers, including round bars, hexagonal bars, square bars, rectangular bars, thin flat strip, and thicker sheets or plates. Other designs call for use of commercially available shapes having good strength and stiffness properties in bending or torsion. Appendix 15 gives 19 tables of dimensional data for several kinds of commercially available shapes:

1. Angles or L-shapes
2. Channels or C-shapes

3. I-beam shapes

4. Hollow tubing: round, pipe, square, and rectangular

Footnotes to the tables suggest suppliers offering the given shapes. You should go to companies' Internet sites to determine actual materials offered.

Suppliers of such shapes typically offer only selected types of materials in a few conditions as stock products. Table 2–6 lists some of the more commonly used materials with their properties, for carbons and alloy steels, stainless steels, and aluminum alloys. These data will be useful to you for design projects where you are asked to specify shape, size, and material for a component. While it is possible to obtain such products in other materials, it is usually less costly to specify standard alloys.

TABLE 2–6 Common Metals and Alloys Used for Commercially Available Rolled, Welded, and Extruded Shapes

Ref.	Material and alloy	Shapes	Tensile strength		Yield strength	
			(ksi)	(MPa)	(ksi)	(MPa)
Carbon and alloy steels						
a	ASTM A36	Rolled shapes; S-beams, angles, channels, plates, bars	58	400	36	248
b	ASTM A500 GR-B	Tubing; round, square, rectangular	58	400	42	290
c	ASTM A992	Rolled W-shapes	65	448	50	345
d	1018 HR	Bars, hexagons, squares, rectangles	55	379	40	276
e	1018 CD	Bars, hexagons, squares, rectangles	70	483	60	414
f	1045 HR	Bars, hexagons, squares, rectangles	90	621	55	379
g	1045 CD	Bars, hexagons, squares, rectangles	90	621	75	517
h	1117 HR	Bars, hexagons, squares, rectangles	65	448	40	276
i	1117 CD	Bars, hexagons, squares, rectangles	80	552	65	448
j	1141 HR	Bars, hexagons, squares, rectangles	95	655	55	379
k	1141 CD	Bars, hexagons, squares, rectangles	100	690	85	586
l	4140 Annealed	Bars, hexagons, squares, rectangles	95	655	54	372
m	8620 HR	Bars, hexagons, squares, rectangles	80	552	60	414
n	8620 CD	Bars, hexagons, squares, rectangles	95	655	80	552
Stainless steels						
o	304 Annealed	Bars, hexagons, squares, rectangles, tubing (round, square, rectangular)	85	586	35	241
p	316 Annealed	Bars, hexagons, squares, rectangles, tubing (round, square, rectangular)	80	552	30	207
Aluminum						
q	2011-T3	Bars, hexagons, squares, rectangles	55	379	43	296
r	2017-T4	Bars, hexagons, squares, rectangles	62	427	40	276
s	2024-T4	Bars, hexagons, squares, rectangles	68	469	47	324
t	3003-H14	Sheet, plate, tubing (round, square, rectangular)	22	152	21	145
u	6061-T6	Bars, hexagons, squares, rectangles, tubing (round, square, rectangular), extruded structural shapes	45	310	40	276
v	6063-T6	Bars, hexagons, squares, rectangles, tubing (round, square, rectangular), extruded structural shapes	35	241	31	214

Notes: Properties are typical but not guaranteed. Other materials are available.
 Obtain detailed information from vendors.
 All listed alloys are ductile with percent elongation >10%.

Structural shapes (L-shapes, C-shapes, and I-beam shapes) used for building construction and some large-scale manufacturing equipment are typically made from structural steel classified by ASTM designations, described in more detail in Section 2–8.

Steel and Aluminum Designations from Different Parts of the World. While this book uses material designations from the United States and other parts of North America,

users of this book may work in any part of the world. Producers of commercially available shapes in other countries will use material designations developed for their own industries and are typically different from U.S. designations. However, many will make alloys that are compositionally similar to U.S. alloys. Table 2–7 lists some common materials and their designations in Germany, the United Kingdom (UK–England), other parts of the European Union (sometimes called Euronorm or EN), Japan, and China.

TABLE 2–7 Comparisons of Designations for Steels and Aluminums from Different Parts of the World

Ref.	USA	European Union (Euronorm, EN)	Germany (W-Number) ¹	(DIN)	UK (BS)	Japan (JIS)	China (GB)
(AISI, SAE, ASTM)		Carbon Steels					
a	A36	S235JRG2	1.0122	S235JRG2	S235JRG2	—	Q235B
b	1015	C15	1.0401	C15	080A15	S15Ck	699-15
c	1018	C18D	1.1141	CK15	040A15	S15	ML3-715
d	1045	C45	1.0503	C45	060A47	S45C	699-45
(AISI, SAE)		Alloy Steels					
e	1213	11SMn30	1.0715	11SMn30	230M07	SUM22	—
f	4130	25CrMo4	1.7218	25CrM04	708A30	SCM 420	ML30CrMoA
g	4140	42CrMo4	1.7225	42CrMo4	708A42	SCM 440H	ML42CrMo
h	4340	34CrNiM06	1.6582	34CrNiM06	817M40	SNCM 447	ML40CrNiMoA
i	6150	50CrV4	1.7222	50CrV4	735A50	SUP10	50CrVA
j	8620	20NiCrMo2-2	1.6523	20NiCrMo2	805M20	SNCM 220 (H)	20CrNiMo
(AISI, SAE)		Stainless Steels					
k	304	X2CrNi18-10	1.4301	X5CrNi18-10	304S 18	SUS 304	0Cr18Ni9
l	316	X5CrNiMo17-12-2	1.4401	X5CrNiMo17-12-2	316S 29	SUS 316	0Cr17Ni12Mo2
m	321	X6CrNiTi18-10	1.4541	X6CrNiTi18-10	321S 31	SUS321	0Cr18Ni10Ti
n	430	X6Cr17	1.4016	X6Cr17	430S 17	SUS430	ML1Cr17
o	17-4PH	X5CrNiCuNb17-4-4	1.4542	X5CrNiCuNb17-4-4	17Cr4Ni	SUS630	0Cr17Ni4Cu4Nb
Aluminum Association		Aluminum Alloys					
p	1100	Al99.0Cu	—	—	—	—	—
q	2014	AlCu4SiMg	3.1255	AlCuSiMn	L.93, L.94	—	—
r	2024	AlCu4Mg1	3.1355	AlCuMg2	L.97, L.98	—	—
s	6061	AlMg1SiCu	—	—	H20	—	—
t	6063	AlMgSi	3.3206	AlMgSi0.5	H19	—	—
u	7075	AlZn6MgCu	3.4365	AlZnMgCu1.5	L.95, L.96	—	—

Notes: ¹Werkstoff Number

The given examples are a small sample of the thousands of available alloys, heat treatments, and formulations. Exact comparisons are not practical. Obtain data from vendors for designations and properties before use.

Sources: Parker Steel Company, Toledo, Ohio.

Steel Strip Company, United Kingdom.

Key to Metals North America, St. Louis, Missouri.

All Metals & Forge, Parsippany, NJ.

Aluminum Association, Arlington, VA.

2-4 VARIABILITY OF MATERIAL PROPERTIES DATA

Tables of data such as those shown in Appendices 3 through 13 normally report single values for the strength, the modulus of elasticity (stiffness), or the percent elongation (ductility) of a particular material at a particular condition created by heat treatment or by the manner in which it was formed. It is important for you to understand the limitations of such data in making design decisions. You should seek information about the bases for the reported data.

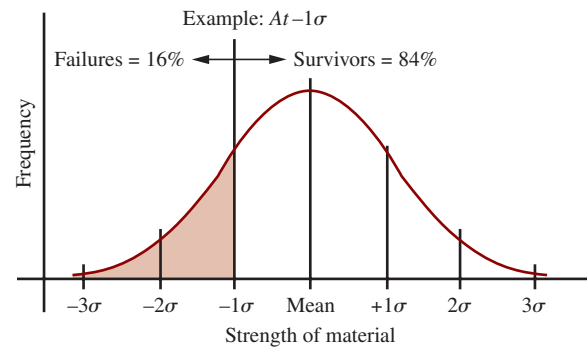
Some tables of data report *guaranteed minimum values* for tensile strength, yield strength, and other values. This might be the case when you are using data obtained from a particular supplier. With such data, you should feel confident that the material that actually goes into your product has at least the reported strength. The supplier should be able to provide actual test data and statistical analyses used to determine the reported minimum strengths. Alternatively, you could arrange to have the actual materials to be used in a project tested to determine their minimum strength values. Such tests are costly, but they may be justified in critical designs.

Other tables of data report *typical values* for material properties. Thus, most batches of material (greater than 50%) delivered will have the stated values or greater. However, about 50% will have lower values, and this fact will affect your confidence in specifying a particular material and heat treatment if strength is critical. In such cases, you are advised to use higher than average design factors in your calculations of allowable (design) strength. (See Chapter 5.)

Using the guaranteed minimum values for strength in design decisions would be the safest approach. However, it is very conservative because most of the material actually delivered would have strengths significantly greater than the listed values.

One way to make the design more favorable is to acquire data for the statistical distribution of strength values taken for many samples. Then applications of probability theories can be used to specify suitable conditions for the material with a reasonable degree of confidence that the parts will perform according to specifications. Figure 2-11 illustrates some of the basic concepts of statistical distribution. The variation of strength over the entire population of samples is often assumed to have a normal distribution around some mean or average value. If you used a strength value that is one standard deviation (1σ) below the mean, 84% of the products would survive. At two standard deviations, greater than 97% would survive; at three standard deviations, more than 99.8%; and at four standard deviations, more than 99.99%.

As the designer, you must carefully judge the reliability of the data that you use. Ultimately, you should evaluate the reliability of the final product by considering the actual variations in material properties, the manufacturing considerations that may affect



Assuming normal distribution of strength:

Stress level	% Surviving
Mean	50
-1σ	84
-2σ	97
-3σ	99.8
-4σ	99.99

FIGURE 2-11 Normal statistical distribution of material strength

performance, and the interactions of various components with each other. There is more discussion on this point in Chapter 5.

2-5 CARBON AND ALLOY STEEL

Steel is possibly the most widely used material for machine elements because of its properties of high strength, high stiffness, durability, and relative ease of fabrication. Many types of steels are available. This section will discuss the methods used for designating steels and will describe the most frequently used types.

The term *steel* refers to an alloy of iron, carbon, manganese, and one or more other significant elements. Carbon has a very strong effect on the strength, hardness, and ductility of any steel alloy. The other elements affect hardenability, toughness, corrosion resistance, machinability, and strength retention at high temperatures. The primary alloying elements present in the various alloy steels are sulfur, phosphorus, silicon, nickel, chromium, molybdenum, and vanadium.

Designation Systems

As mentioned in Section 2-3, the designation systems for steels are managed by either SAE International or ASTM International. The wrought or rolled steels typically used for machine elements are most likely to be selected from SAE grades. Structural steels and many cast metals carry ASTM grade designations. It was pointed out earlier that, until the recent past, both SAE and AISI issued wrought or rolled steel designations that were substantially identical, but AISI no longer performs this function. However, it is likely that much published literature and commercial product information will continue to use the term *AISI* until the new systems are phased in.